# From the Pulpit to the People: Experimental Evidence on the Impact of Friday Sermons

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**Abstract**

I explore the role of religious institutions in shaping behavior by experimentally varying the content of Friday sermons - one of the most widespread forms of in-person religious persuasion in the Muslim world - in 309 mosques in Pakistan. Working with the Punjab Auqaf Department that administers state religious properties, I randomize mosques to deliver sermons on women's rights, neighbors' rights, or a control condition, with additional variation in whether the arguments used are religious or secular. I find sizable impacts of women's rights sermons (0.23 SD) and neighbors' rights sermons (0.19 SD) on men's behavior as measured through lab-in-field games, field behavioral measures, and survey responses 2-4 weeks after treatments. I find suggestive evidence that effects extend indirectly to wives, who do not hear sermons themselves and yet report more prosocial beliefs. The effectiveness of religious leaders' messaging derives at least partly from the types of arguments made as sermon content is more persuasive when grounded in religious arguments for women's rights but in secular arguments for neighbors' rights. Examining equilibrium dynamics, I document supply-side persistence in prosociality of sermon content alongside demand-side backlash as evidenced by a 20% decline in attendance numbers at mosques that received sermons on women's rights that persists in the long run. A two-sided Bayesian framework with information avoidance, leader learning, and conservative religious priors constraining updating from secular signals rationalizes these findings.

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# 1 Introduction

Long before mass media, persuasion at scale happened via the pulpit. Today, religious sermons are estimated to reach hundreds of millions of listeners weekly, making them some of the most widespread and regular channels of in-person persuasion in the world.[1] Whether this content shapes what people believe and do, or only reflects these things, is a fundamental question in social science (Iannaccone, 1998; Barro and McCleary, 2003; Becker and Woessmann, 2009; Iyer, 2016). If religious messaging is indeed persuasive, even less is known about why: is its effectiveness tied to the authority of religious leaders, or is the religious framing of such content key? Causal evidence on these questions is sparse, for the simple reason that how people choose congregations and how religious leaders decide what to say are non-random.

Utilizing a novel partnership with the Punjab Government's Auqaf and Religious Affairs Department, this paper experimentally varies sermon content in the universe of 309 state mosques in Punjab, Pakistan. It studies the causal impact of the Friday sermon, one of the most central religious institutions in Muslim societies. The Friday sermon precedes the Friday congregational prayer which is obligatory for adult Muslim men, and is often considered to be the primary formal channel through which Muslim religious leaders - the imams - interpret, frame, and transmit religious teaching to their communities en masse (Gaffney, 1994). On the other hand, women, who are largely absent from these congregations, can only be impacted by the content indirectly, by what men in their household say or how they behave. The importance of Friday sermons has led many states (e.g. Egypt, Turkey, and Saudi Arabia) to centrally mandate sermon content (Ozdalga, 2022); in many others, including Pakistan, imams typically retain considerable discretion over what to say, though governments occasionally utilize them for information campaigns. A causal evaluation of Friday sermons thus offers a unique opportunity to study belief formation in the presence of others, the formation and intra-household transmission of social norms, and the potential for sermons to serve as policy tools for information dissemination.

The Punjab Auqaf Department employs the imams in these mosques who lead the Friday congregational prayer for approximately 116,000 mosque-goers each week.[2] The unique institutional setting makes the experiment possible because the Department holds the authority to mandate sermon content but does not exercise it in normal weeks, allowing me to randomize the content imams deliver while retaining a natural control group of business-as-usual sermons delivered without

[1] While there is little data on the total number of individuals who attend sermons regularly, it is estimated that 40% of adults worldwide attend religious services at least weekly (Kramer and Fahmy, 2018). This number is generally higher in many Muslim-majority nations (e.g. it is 59% in Pakistan). In the sample from Punjab, Pakistan surveyed for this study, 90% of those who attended the Friday congregational prayer reported listening to at least 10 minutes of their imam's sermon the previous Friday.

[2] Estimates of weekly Friday prayer attendance are derived from baseline surveys of each mosque.

any state input.

I randomize mosques to receive sermons on women's rights, neighbors' rights, or a control condition in which imams deliver their usual content, cross-randomizing whether treatment sermons are framed in fully religious or partially secular terms. The framing variation tests whether religious leaders are persuasive regardless of what they say or whether their messages must be grounded in religious arguments. The choice of topics, in turn, captures two domains in which religion pulls in opposite directions. Religious interpretations in Pakistan are often deeply conservative on women's rights, while mainstream interpretations of Islam emphasize kindness toward neighbors (Weiss, 2014), allowing me to test whether religious framing is more persuasive when it reinforces or challenges prevailing religious views. Furthermore, the women's rights topic carries substantial policy significance in a setting where religious teachings are frequently invoked to justify restrictions on women's participation in social life (Shaheed, 2010).

Each treatment takes the form of two distinct sermon scripts that imams are asked to deliver over consecutive Friday sermons. The women's rights sermons emphasize kindness toward women and the importance of women's education, employment, and social participation, and conclude with a call for men to discuss aspirations for their daughters with their wives. On the other hand, the neighbors' rights sermons emphasize kindness toward neighbors, the obligation to help neighbors in need, and the importance of community cleanliness, concluding with a call for men to spend an hour cleaning their neighborhood in the coming week. Within each topic, the religiously framed versions ground these arguments entirely in the Quran, the life of the Prophet Muhammad, and early Islamic history - citing, for instance, the Prophet's wives as teachers of senior companions, or presenting the duty to lend money and send food to neighbors in need as religious obligations - while the partially secular versions present the last quarter of each script in secular terms, arguing for example that educated women can better support their children's education and supplement household income, and that good relations with neighbors allow households to share resources.

To evaluate compliance, I record and transcribe sermons in both treatment and control mosques and analyze this rich text data using ratings given to content by a custom LLM trained for the analysis of sermon content in the local language. In doing so, I document a strong first stage: imams followed the assigned scripts relatively closely, with an average adherence rating of 7/10 that does not differ across treatment arms. The recordings also allow me to characterize what control imams, who delivered their business-as-usual sermons free of any state input, discuss: control sermons almost never discuss rights, focus far more on purely religious topics, are more conservative, and are more likely to contain hate speech.

I find that sermons have sizable impacts on behavior as measured through my primary preregistered outcomes 2-4 weeks after conclusion of treatments. Men who attended mosques where sermons on women's rights were delivered allocated 33% more money to a fund that helps women

find employment, were 19.5% more likely to consent to having their wife surveyed, and 11% more likely to sign a petition to support women's political participation. Men who prayed in mosques with sermons on neighbors' rights donated 19% more to poor neighbors but there was no change in enumerator-evaluated community cleanliness. When summarized in an index, the women's rights and neighbors' rights sermons produced behavioral responses of 0.23 SD and 0.19 SD respectively.[3] My reliance on behavioral measures as primary outcomes, together with the fact that men heard the sermons from their own imams at their home mosques months after being recruited at baseline, and that the sermons they received were never revealed to be part of the study, limits the scope for experimenter demand effects.[4] Results are also robust to controlling for the baseline value of the outcome where available, and I find no evidence of differential attrition, significant spillovers on other men, or social desirability bias.

I find suggestive evidence that effects extend downstream to wives, who are not present for the Friday prayer and can only be affected through intra-household transmission, by what their husbands say or how they behave in the household. Surveyed by phone conditional on the husband's consent, wives of men who received women's rights sermons report 0.19 SD more prosocial beliefs about women. Wives also perceive changes in their husbands' beliefs and behavior within the household, reporting that their husbands are more supportive of their daughters' education but also that they spend less time talking to family. While the downstream effects on women may reflect a selected sample of wives, the magnitude of effects and several tests for selection suggest that the sermon treatments had real indirect effects on women.

Next, I evaluate the impact of framing, whether the effects I observe depend on the content being framed in religious language or whether anything the imam says is persuasive. Examining treatment effects across my framing arms reveals a striking asymmetry in the effects on men: for women's rights, religiously framed sermons produce effects of 0.33 SD, more than double the 0.14 SD effect of secularly framed sermons, and the difference between the two treatment arms is statistically significant. The pattern reverses, however, for sermons on neighbors' rights: secularly framed sermons produce a larger effect of 0.25 SD, while the effect of religiously framed sermons is smaller at 0.14 SD. These patterns suggest that framing matters and heterogeneously by topic, with similar prosocial content being more effective when grounded in religious arguments for women's rights but in secular arguments for neighbors' rights.

Delivering treatments in their natural setting at scale allows me to evaluate equilibrium dynamics at the mosque level, which I turn to next. On the supply side, I evaluate how my treatments

[3]These behavioral effects are accompanied by smaller shifts in self-reported beliefs: men exposed to women's rights sermons report a 0.18 SD increase in support for women's education and a 0.11 SD increase in a pooled women's rights belief index, while I find no statistically significant change in beliefs about neighbors.

[4]During piloting, fewer than 5% of respondents who were asked directly at the end of the survey what they believed the study was about were able to identify that it concerned sermons.

endogenously changed the sermons imams delivered the first week after treatments had concluded and they were free to say whatever they wished. Imams who delivered treatment sermons continue to discuss rights in their subsequent sermons and their content becomes significantly less conservative and less likely to contain hate speech in the short run.

On the demand side, I look at impacts on mosque-level congregation numbers and find that the positive impacts of sermons I have documented thus far came at a cost. Mosques in which the religiously framed women's rights sermons were delivered, where behavioral effects were largest, experienced a 20% decline in total attendance as measured directly by enumerators, while I observe no similar attendance effects of sermons on neighbors' rights. I interpret this as evidence of backlash: mosque-goers vote with their feet against the highly prosocial content on women. I also look for evidence of additional negative consequences from a policy perspective but find that the treatments neither reduce imams' satisfaction with their government jobs nor lead to domain expansion where imams believe they should have greater say over non-religious issues such as politics.

I make sense of these results with a model of this market for religious content. On the demand side, I model congregants as deciding whether to attend their mosque each Friday and then, after hearing the sermon, how to act. They value both a religious and secular objective, conforming to God's will and doing well in life, each of which is uncertain. Sermons provide signals about these optimal actions depending on their framing, and congregants update as Bayesians, with signals being more persuasive when the relevant priors are less prosocial. Crucially, congregants attend only if the value of the sermon exceeds the expected cost of changing behavior in response to a signal that conflicts with priors, so highly prosocial content on high-stakes topics with anti-social priors can drive attendance down. On the supply side, imams choose sermon content to balance the utility from teaching what they themselves believe ("preaching" utility) against the "pandering" utility derived from teaching content close to congregants' priors (which they learn about from attendance movements).

The model rationalizes both the framing asymmetry, whereby religious framing produces larger effects for women's rights but secular framing is more effective for neighbors' rights, as well as the backlash being limited to the religiously framed women's rights sermons, through the structure of congregants' religious and secular priors; in the model, signals are most persuasive exactly where they conflict with less prosocial priors, and the same conflict simultaneously produces the greatest attendance losses. The structure of priors in my data matches this logic: using baseline data, I construct indices measuring individuals' religious and secular priors and find that congregants' religious priors are less prosocial than their secular priors for women's rights, while the reverse holds for neighbors' rights. No attendance decline appears for neighbors' rights because secular priors about neighbors are still relatively prosocial and because revising one's behavior toward neighbors is likely to be less costly than revising one's behavior toward women in the household.

My framework attributes the persistence of prosocial content to a shift in imams' own beliefs. I test this by conducting detailed surveys of imams, administering them the same questions asked of the men in my sample. I find that delivering the sermons made imams more prosocial themselves, consistent with past work documenting self-persuasion when individuals advocate for a position in public (Schwardmann, Tripodi and van der Weele, 2022). The effects on imams are largest in the religiously framed women's rights arm, exactly the arm in which subsequent sermon content became most prosocial on women. In the long run, however, my model predicts that content should revert: imams observe the attendance movements at their mosques, and sustained losses with the "pandering" term in the utility function pull content back toward congregants' preferences. I return to my study mosques 6 months to 1 year after treatments have concluded to evaluate long term impacts on sermon content and attendance. I find that prosocial content has mostly decayed in women's rights mosques while attendance effects have persisted, consistent with the model and information frictions preventing congregants from observing content in their original mosque.

This standard Bayesian framework, however, only partially explains the treatment effects across topics: the secular treatment effect for women's rights is small despite relatively low secular priors, while the secular treatment effect for neighbors' rights is large despite high secular priors. To resolve this puzzle, I extend the model to allow religious priors to "gate" secular persuasion: when individuals' religious priors are too low, they place little weight on the secular objective and under-respond to secular signals, while high religious priors license engagement with secular arguments. The extended model's key new prediction - that secular treatment effects should be increasing in religious priors, in stark contrast to the standard Bayesian case of no cross-domain interaction - finds support in the data: the secular treatment effect for women's rights nearly triples for respondents with above-median religious priors, a difference significant at the 10% level. Besides reconciling the patterns in my data, this exercise highlights the centrality of religious priors and may help explain why so many secular campaigns aimed at changing men's attitudes about gender that left religious beliefs unaddressed have produced only modest effects.

Finally, I explore additional mechanisms that may be responsible for driving my effects. The first is the role of sharing sermon content with others, which I evaluate using an information-sharing cross-randomization whereby half of all treatment sermons concluded with a plea from the imam to share what was heard with wives. The expected direction of effects is theoretically ambiguous: increased information sharing could reinforce effects if sharing sermon content and the threat of cognitive dissonance leads to behavioral change à la the social psychology literature, but it could also dampen effects if sharing information designed to increase women's autonomy and well-being shifts household bargaining dynamics in ways men prefer to avoid (Aronson, Fried and Stone (1992); Cullen et al. (2024)). In practice, I find relatively small differences in treatment effects across the information sharing and no sharing treatment arms and cannot reject the null hypothesis of no

difference.

Another potential mechanism is social norms: because sermons are delivered publicly to large congregations, each listener receives a signal not only about religious doctrine but also about what others believe, so effects may be operating through second-order beliefs. However, I find little support for this channel. Decomposing endline beliefs into personal beliefs, beliefs about social norms, and religious beliefs reveals that the treatments move personal and religious beliefs while leaving perceived norms unchanged. Moreover, treatment effects do not vary meaningfully by congregation size or whether the mosques are located in residential areas, variables that my randomization was stratified by, even though a norms channel would predict larger effects in bigger congregations of neighbors.

This paper contributes to several literatures. First, it provides some of the first experimental evidence on the causal effects of religious messaging delivered through regular institutional channels, contributing to the broad literature evaluating the effect of religion on various economic outcomes (Iannaccone (1998); McCleary and Barro (2006); Clingingsmith, Khwaja and Kremer (2009); Campante and Yanagizawa-Drott (2015); Wang (2021)). It is only the second paper, to the best of my knowledge, after Bryan, Choi and Karlan (2020) to be able to randomize religion in real-world settings. I randomize religious messaging conditional on exposure as opposed to religious exposure itself, but my findings largely complement those of Bryan, Choi and Karlan (2020) in documenting the importance of religion in shaping economic and social outcomes.

My paper also contributes to a very recent literature evaluating the impact of religious leaders (Prummer (2019); Buccione (2023); Campbell, Sethuraman and Steffen (2024); Avdeenko et al. (2025)). It complements the key finding of this literature that religious leaders matter by studying a key channel through which religious leaders reach their communities worldwide: in-person sermons. The closest paper to mine is recent work by Sarkar, Elverdin and Lucek (2026), who study the effects of Christian sermons in Colombia by randomizing respondents to audio recordings of sermon content during a survey experiment, and find that Evangelical Pentecostal messages emphasizing women's agency and competence produce only modest effects on attitudes. I complement this work by studying the key institution of the Friday sermon in Islam, randomizing sermons directly in their natural congregational setting, and by documenting large impacts of sermons emphasizing women's agency on costly measures of men's behavior towards women.

Second, this paper contributes to a large literature on persuasion that explores the effects of information provision in various contexts (DellaVigna and Gentzkow (2010); Haaland, Roth and Wohlfart (2023); Yanagizawa-Drott (2014); La Ferrara, Chong and Duryea (2012)). The information-provision literature has established that beliefs update in response to new information, but typically studies one-shot, factual treatments delivered often in artificial settings by researchers and is often silent on the process of belief formation itself. There are some notable exceptions to

this: Yanagizawa-Drott (2014) studies the role of radio in inciting violence in Rwanda, La Ferrara, Chong and Duryea (2012) examines entertainment television's effects on fertility decisions in Brazil, and DellaVigna and Kaplan (2007) evaluates the persuasive effects of political media in the United States. My paper contributes to this literature by evaluating the persuasive impacts of in-person congregational sermons delivered by highly trusted religious leaders. Sermons are a qualitatively different persuasion technology: they potentially combine religious authority, repetition, and social reinforcement in a public congregational setting, allowing me to study how the social and religious dimensions of information provision interact to influence belief formation. Furthermore, my finding that priors in one domain can influence updating from signals in a different domain offers a refinement to the classic Bayesian models that are standard in this literature.

Finally, by randomizing sermon content to focus on women's rights and evaluating impacts on men's beliefs and behavior, this paper speaks to a growing body of work that studies interventions aimed at shifting gender attitudes, including media campaigns (Jensen and Oster (2009); La Ferrara, Chong and Duryea (2012)), educational programs (Dhar, Jain and Jayachandran, 2022), and economic interventions (Field et al., 2021). This work, taken together, finds that men's beliefs on gender can be sticky, sometimes documenting backlash in response to campaigns aimed at improving women's autonomy and economic participation (Cullen et al., 2024; La Ferrara and Yanagizawa-Drott, 2026). Surprisingly, this literature is mostly silent on the potential for religion and religious leaders to shift beliefs about gender. This is despite the fact that conservative religious priors may themselves be a key reason why gender norms are so resistant to change, particularly in Muslim-majority societies where religious teachings are frequently invoked to justify restrictions on women's education, employment, and social participation. My work finds that religious leaders can be especially effective at shifting men's beliefs on gender, producing behavioral changes of 0.23 SD. Moreover, the finding that framing matters, that identical prosocial content on women's rights is more powerful when grounded in religious rather than secular arguments, and that secular information can be ineffective when religious beliefs are left unchanged, reinforces the centrality of religion as both the key bottleneck and a potential lever for changing gender norms, having important policy implications for highly religious settings.

The remainder of the paper is organized as follows. Section 2 provides information on the context and institutional setting. Section 3 provides detail on my experimental design and data. Section 4 presents my empirical results. Section 5 presents my theoretical framework and uses it to interpret my results. Finally, Section 6 concludes.

# 2 Context and Institutional Setting

## 2.1 The Friday Prayer and Sermon in Islam

The Friday congregational prayer, known as *Salat al-Jumu'ah* in Arabic, is the most important institution of collective worship in Islam. It is performed at mid-day every Friday and all major Sunni sects consider attendance obligatory for adult Muslim men.[5] The importance of the ritual has meant that in many Muslim countries, Friday is part of the official weekend while in others, Fridays are half-days for schools and businesses. Unlike the five daily prayers which may be performed individually, the Friday prayer is inherently congregational and must be performed in a group, most often inside a mosque. This congregational requirement transforms the prayer from a private religious practice into a recurring social institution, bringing together men from a given neighborhood at the same time and place each week. Women do not have the same religious obligation to offer the prayer in a congregational setting as men, but may join the congregation if the mosque has a separate prayer section for women.

The Friday prayer is preceded by the Friday sermon (*Khutbah* in Arabic) which is delivered by the imam, the primary religious leader in Muslim societies. Imams utilize sermons as platforms for religious teaching through discussions of theology, morality, correct ritual practice, as well as Islamic history, but sermon content often extends beyond just these subjects to social issues facing local communities as well as traditionally secular topics such as politics, economics, and health (Gaffney, 1994). The tradition traces directly to the Prophet Muhammad, who delivered the earliest Friday sermons himself and used sermons both to preach and as tools for political mobilization (Borthwick, 1967). Traditional Islamic law required the Friday sermon to be delivered entirely in Arabic and historically, Friday congregational prayers were restricted to central mosques in major urban centers, often limited to one per city. Today, sermons in non-Arabic speaking countries are delivered primarily in the local language and are held in mosques of all sizes which has vastly expanded the reach of the institution.

The scale of the Friday sermon as a communication channel is difficult to overstate. While no precise numbers are available on the number of individuals that listen to the Friday sermon regularly, an estimated 40% of adults in Muslim-majority countries attend religious services at least weekly, and in many countries the share is considerably higher (Kramer and Fahmy, 2018). Presumably, many of these individuals also listen to the Friday sermon; in my own sample from Punjab, Pakistan surveyed for this study, 93% of those who attended the Friday congregational prayer reported listening to at least 10 minutes of their imam's sermon two Fridays ago. This makes

[5]The Quran is explicit about the importance of the Friday prayer: "O believers! When the call to prayer is made on Friday, then proceed diligently to the remembrance of Allah and leave off your business (Surah Al-Jumu'ah, 62:9-10)."

the Friday sermon one of the most widespread and regular channels of in-person persuasion in the world, reaching hundreds of millions of people each week.

Recognizing the reach of the Friday sermon and its potential to mold public opinion, several Muslim-majority states have sought to centralize control over sermon content. For instance, in Turkey, the Presidency of Religious Affairs (Diyanet) has sought to standardize Friday sermon content since the early years of the Republic, and today distributes a single centrally-written sermon each week that is often read verbatim in mosques across the country (Ozdalga, 2022). In Egypt, the Ministry of Awqaf writes and disseminates official Friday sermons, monitors compliance in major mosques, and has been known to discipline imams who deviate from the approved text. These efforts have not gone uncontested. When Egypt moved to mandate pre-written sermons in 2016, congregants in some mosques reportedly shouted down imams reading from the official scripts, and a council of senior scholars publicly rejected the policy, arguing that standardized sermons would erode imams' capacity for independent religious reasoning (Associated Press, 2016). More recently in 2024, Morocco's announcement that it would standardize the Friday sermon drew widespread backlash and caused the Ministry of Endowments and Islamic Affairs to backtrack and clarify that the decision to unify Friday sermons was temporary and would preserve the autonomy of imams (Faouzi, 2024). These examples illustrate that governments across the Muslim world view the Friday sermon as a powerful policy instrument and that the question of who controls sermon content continues to be an important and contested policy issue.

Given the significance of the Friday sermon in Muslim societies, a large literature outside of economics has emerged that studies them. In some of the earliest work on the subject, Borthwick (1967) analyzed radio sermons in the Middle East to argue that ruling elites utilize Friday sermons as a channel of political communication to transmit nationalist messages, with preachers acting as mediators who clothe state policy in the "legitimacy myths of the Islamically oriented masses." In an Anthropology study of preachers in rural Jordan, Antoun (1989) characterized the imam as a "culture broker" responsible for applying Islamic teachings to the everyday social concerns of his congregation. Based on fieldwork in Egypt, Gaffney (1994) argued that the sermon is the preeminent medium through which the message of Islam reaches ordinary believers, and traced how preachers convert religious authority into political influence. More recently, Aksoy (2023) uses machine learning to analyze all Friday sermons read to citizens of Turkey between 2015 and 2021 and finds that their content responds to current events and also predicts subsequent public discussion on social media. Yet, this work is descriptive or correlational. Given the difficulty in randomizing sermons, we still lack sound empirical evidence on the causal impact of this important religious and social institution on individuals' beliefs and behavior.

## 2.2 Religion and Gender in Punjab, Pakistan

Punjab, Pakistan's most populous province home to over 127.6 million people, is overwhelmingly Muslim: approximately 98% of the population identifies as such, with the majority following the Sunni tradition (Pakistan Bureau of Statistics, 2023). Moreover, religious observance is also high; nationally, over 90% of Pakistanis report that religion is very important in their daily lives and 59% of Muslims report attending their mosque at least weekly (Pew Research Center, 2012). At the center of this religious life is the imam, who occupies a position of considerable religious and social authority in Pakistan. For many Pakistanis, the relationship with the imam begins in childhood: most children receive their earliest religious instruction through mosque-based classes led by the imam, establishing him as a trusted figure from a young age. As adults, the imam continues to serve a central role and acts as a local leader by offering religious guidance, leading congregational prayers, conducting funeral and wedding rites, and helping fund-raise for and organize various community causes among other responsibilities. While formal religious instruction largely ends after childhood, the Friday sermon remains, often making it the primary channel through which many adults continue to receive religious guidance over the course of their lives.

The primary structure out of which the imam operates is the mosque. In Muslim societies, the mosque acts not only as a place of worship and religious instruction but also as a hub for community life, serving to strengthen social networks, aid economic activity, and even act as a space for conflict resolution.[6] Utilization of the mosque for religious and social purposes, however, is largely limited to men. The audience for the Friday prayer and sermon is almost exclusively male, reflecting both the religious obligation - limited only to men - and prevailing social norms in Pakistan that limit women's mobility and social participation.[7] The gender-segregated institutional structure, however, is both a feature of the setting and a motivation for the study: any effect of sermon content on women must operate indirectly, through intra-household transmission of information from husbands to wives, making the Friday sermon a useful setting for studying how information, particularly religious information, flows within households.

The role of religion in influencing beliefs about gender is especially pertinent in my setting which suffers from some of the starkest gender disparities in the world. The World Economic Forum ranked Pakistan last of 148 countries in its Global Gender Gap Report in 2025, with large deficits in economic and political participation of women. Female labor force participation stands at roughly 24%, far below the world average of 49% and low even relative to South Asia (World

[6]The important role of the mosque in all parts of community life dates back to the time of the Prophet Muhammad, whose mosque in Madinah served not only as a place of worship but also as a court for settling disputes, a center for distributing public revenue, a place to receive foreign delegations, and a shelter for the poor (Omer, 2013).

[7]In theory, women are allowed to participate in congregational prayer at mosques if there is a separate section for women but the lack of such spaces as well as the fact that there is no similar religious obligation on women to pray in congregations makes this rare in practice.

Bank, 2025). Social norms restricting women's mobility, access to public spaces, and interactions with non-relatives are pervasive and represent binding constraints on women's economic and social opportunities.

Moreover, religious teachings are frequently invoked to justify such restrictions on women. Conservative interpretations of Islamic texts are routinely cited to argue that women's primary roles are domestic, that gender mixing is impermissible, or that women's mobility should be restricted (Weiss, 2014). At the same time, Islamic doctrine and history contain substantial support for women's participation in social life, with the Prophet Muhammad's first wife, Khadijah, being a successful businesswoman and numerous prophetic sayings emphasizing the obligation of seeking knowledge for both men and women. The gap between what the Islamic scriptural and historical tradition contains and what congregants usually hear from their imams on the topic of women's rights is important for the design of the study.[8] If imams rarely address women's rights in their sermons or if the usual interpretation is restrictive, then experimentally varying sermon content to foreground the prosocial elements of Islam offers an opportunity to test whether religious messaging can shift deeply held beliefs about gender.

## 2.3 The Auqaf Department and Study Mosques

The study is implemented through an official partnership with the Punjab Government's Auqaf and Religious Affairs Department. The Auqaf Department is a provincial government body responsible for the administration of *Auqaf* properties which are Islamic charitable endowments known as *Waqf*, plural *Auqaf*, that include mosques, shrines, and associated land and other assets. The department was established in 1960 under the West Pakistan Waqf Properties Ordinance of 1959, which nationalized the administration of prominent religious endowments in the country. Several historians have chronicled the history of the department, arguing that its goal was to increase state control over important religious sites whose hereditary saints (*pirs*) were locally influential and a potential threat to state power, extract the large revenues these sites generated, and push the nation-building ideology of a "Singular Islam," weakening the pluralistic local practices historically associated with many of these sites (Malik, 1996; Ibad, 2019). In informal conversations, department bureaucrats noted that the most important reason sites are likely to be acquired today is in the aftermath of a dispute regarding ownership. Nationalization is nonetheless the exception rather than

[8] Conservative religious discourse in Pakistan often emphasizes injunctions regarding veiling and gender segregation, drawing selectively on Quranic verses and prophetic traditions to argue against women's presence in public life. It is important to note that the sermon scripts used in this study do not challenge these interpretations, but rather draw on well-established verses and historical examples that are squarely within mainstream Sunni Islam but that are typically under-emphasized in regular Friday sermons. The experimental variation thus emphasizes prosocial arguments already present in the tradition rather than introducing novel or heterodox claims, with the scripts being drafted by mainstream religious scholars employed by the Auqaf Department and approved by a committee comprising senior religious leadership.

the rule: Pakistan's first economic census counted roughly 600,000 mosques nationwide (Rana, 2025), with the vast majority of mosques and shrines built, financed, and managed privately by local communities. The department's mandate instead is the maintenance and regulation of a small fraction of these sites, often the most historically and religiously significant ones in the province. The sample utilized for this study is the universe of 309 Auqaf-administered mosques in the Punjab province.[9]

Crucially for this study, the Auqaf Department directly employs the imams who serve in its mosques, making them salaried government employees. Because Auqaf imams are state employees, the department exercises a degree of administrative authority over them that would not exist in privately-managed mosques, making the experimental design of this study possible. The department occasionally does direct imams to deliver content on specific topics, lending institutional precedence to the type of intervention studied here.[10] However, such campaigns are rare, and in ordinary weeks, imams at Auqaf mosques exercise considerable discretion over the content of their Friday sermons, much like imams at privately-managed mosques. This contrasts with the types of centralized sermon regimes discussed earlier, such as those in Egypt or Turkey where sermons are mandated each week, and provides a natural control group in which imams deliver their standard sermons with no state input as well as the opportunity to evaluate how business-as-usual sermons change in the aftermath of a prosocial messaging campaign.

Auqaf mosques are geographically dispersed throughout the province, spanning major urban centers, smaller towns, and rural areas. Their locations are shown on a map of Punjab in Figure A1. Approximately 13% are located in rural areas (the rest are urban), 37% are situated in residential neighborhoods (as opposed to commercial areas), and 25% also have an associated Sufi shrine. Almost all of the mosques in my sample are Sunni, with just one being Shia. According to estimates derived from baseline data, the median mosque is attended by approximately 250 congregants every week for Friday prayer and weekly Friday prayer attendance for all 309 mosques totals approximately 116,000 individuals. The Friday prayer is preceded by a sermon in Urdu, and occasionally Punjabi, and is anywhere from 15-45 minutes long.[11] An example congregation receiving treatment sermons is shown in Figure A2.

It is important to note that Auqaf mosques represent a specific subset of the broader mosque landscape in Punjab. They tend to be older, more historically prominent, and are more likely to be

[9]The Auqaf Department officially notes that it has 437 mosques under its jurisdiction, but when enumerators visited these sites, many had no evidence of government oversight nor any government staff. I also excluded around 20 mosques that were used during the pilot, as well as several key mosques in Lahore, such as Badshahi Mosque and Data Darbar, that attract tens of thousands of worshipers for Friday prayer, making treatment delivery and, particularly, measurement of attendance there difficult.

[10]For instance, in the past, imams have led public health campaigns, assisted with providing flood relief, and delivered messages of communal harmony in the aftermath of sectarian violence.

[11]The average sermon length is 27.9 minutes.

located in dense urban areas or near shrines compared to the hundreds of thousands of privately managed mosques across the province. Thus, we may worry that the congregants who attend Auqaf mosques differ relative to the broader population, especially problematic if attending a state mosque selects for more pro-state individuals. However, at least anecdotally, the vast majority of congregants that we interacted with at these sites tended not to be aware that the mosques were run by the state or even what the Auqaf department was. This is corroborated by my baseline survey, in which trust in state institutions was low and consistent with estimates from other studies with more generalizable samples.[12] Moreover, mosque choice in Pakistan is driven overwhelmingly by proximity: in my baseline data, 74% of respodnents reported that distance was the most important factor in choosing where to pray. This suggests that while there is likely geographic selection, the congregants in my sample are broadly representative of the neighborhoods in which these mosques are located. I return to discussions of generalizability in Section 3.2.

# 3 Experimental Design and Data

This section describes the experimental design and data for this study. Fieldwork started in September 2024 and concluded in July 2026. Piloting was conducted in Summer 2023.[13]

## 3.1 Experimental Design

### 3.1.1 Treatments

The primary treatments utilized for this study are sermon scripts with prosocial sermon content on either women's rights or neighbors' rights. Each treatment consists of two distinct scripts, delivered over consecutive Fridays, with mosques in the pure control group receiving business-as-usual sermon content without any prompting from the government. The treatment sermons were developed by two senior religious scholars employed by the Auqaf Department, and were reviewed and approved by an Auqaf leadership committee consisting of senior imams and department administrative staff to ensure that the content was theologically sound and uncontroversial. The two treatment topics were chosen for their policy relevance - low levels of women's social and economic participation and limited social trust and cooperation are pervasive in Pakistan - and allow me to evaluate the extent to which the domain of imams' messaging influences their effectiveness.

[12]The mean response to the question, "To what extent do you trust state institutions?" was 4.6 on a 0-10 scale, almost identical to the 4.9/10 documented in Acemoglu et al. (2020) for a random sample of rural male household heads in Punjab, Pakistan.

[13]The study was pre-registered at AEARCTR-0012715 and received ethical approval from Harvard University-Area Committee on the Use of Human Subjects with protocol ID IRB24-0495.

Each sermon script emphasizes unique components of women's and neighbors' rights within the Islamic tradition. The first women's rights script discusses the importance of kindness towards women, drawing extensively from the life of the Prophet Muhammad, and ends with the importance of women's education in Islam. The second script focuses on the importance of women's participation in social, economic, and political life, using examples from the Quran as well as early Islamic history. On the other hand, the first neighbors' rights script discusses the importance of neighbors in Islam, emphasizing the obligation to help neighbors in need, including through loans and by sending them food. While the second focuses on the importance of not harming neighbors, the role of the community mosque in building social trust, and community cleanliness as a religious and social obligation. All sermon scripts conclude with a concrete action that congregants are asked to perform in the following week to be able to apply what they have heard: the women's rights sermons ask men to spend time discussing their aspirations for their daughters with their wives, while the neighbors' rights sermons ask men to spend an hour the following week cleaning their neighborhood.

The following excerpts, illustrate the content of the scripts. The women's rights sermons argue, for instance:

> *Dear brothers! In the name of God, the most compassionate, the most merciful... One of the rights of women is that they have been given the right to work... Hazrat Khadijah, the first wife of the Prophet had a business before Islam and after the marriage, the Prophet never prevented her from engaging in it, but on the contrary, appreciated her for helping him out with financial matters.*

Similarly, the neighbors' rights sermons ground their appeals in analogous prophetic teachings:

> *The Holy Quran and the Noble Prophet have given great importance to the relationship of neighborliness in their teachings and guidance... Islam outlines several rights of neighbors. The Prophet said: "He is not a believer who sleeps full while knowing his neighbor is hungry."*

To explore mechanisms, I also cross-randomize two additional dimensions within the treatment arms: the framing of the sermon content and whether imams encourage transmission of sermon content to wives. The framing dimension varies whether the final quarter of each sermon script is presented in fully religious language - grounding arguments solely in Quranic verses, stories from the Prophet Muhammad's life, and Islamic history and law - or in secular language, appealing to modern practical, economic, and social concerns with little religious content.[14] This variation

[14]To illustrate the difference more concretely, the first religiously-framed women's rights script argues, using sayings from the Prophet Muhammad, that seeking knowledge is obligatory for every Muslim and gives the example of the Prophet's wives as teachers of senior companions, while the secular version argues that educated women can better support their children's education, manage household budgets more effectively, and supplement household income. In a

helps evaluate what types of messaging are effective for imams to deliver, the extent to which the imam's persuasive authority is tied to religious argumentation, and also allows me to test the model presented in Section 5, which predicts that treatment effects will depend on how the framing of content interacts with a person's priors. The entire sermon cannot be framed in fully secular language because it would be culturally artificial in my context, so the secular framing arm should be thought of as making the sermon *more* secular rather than fully secular while the religiously-framed sermon is fully religious.[15] The information-sharing dimension varies whether sermons conclude with an explicit plea from the imam asking congregants to share what they heard with their wives, or end without such a plea.[16] The goal of this second cross-randomization is to evaluate whether increased sharing of religious information in the household increases effects for men (e.g. through reinforcement effects or the threat of cognitive dissonance) or attenuates them (e.g. by reducing the bargaining power of men after discussions on women's rights). The control group, absent any treatment scripts, are not cross-randomized to either of these sub-treatments. Full sermon scripts, translated from Urdu into English, associated with each of the treatments arms are provided in Appendix.

This yields a 2x2x2 design within the treatment arms plus a control group, for a total of 9 experimental cells, as summarized in Table A1. Randomization was stratified by mosque size (i.e., Friday prayer attendance tercile) and residential/commercial status of the mosque to allow for clean evaluations of heterogeneity in treatment effects across these key dimensions at endline; for instance, larger mosques may provide stronger signals about social norms but less personal exposure to the imam, while the effect of the neighbors' rights treatment may be reinforced in residential mosques if kinder neighbors following sermons further reinforce treatment effects.

To implement treatments, imams were invited to their respective Auqaf district offices where they received a presentation explaining the study and were given two printed sermon scripts to deliver over the following two consecutive Fridays as close to verbatim as possible. District head imams and senior Auqaf personnel were present at these meetings to address imams' concerns and to lend institutional weight to the request, maximizing the probability of compliance. Control imams were not invited to these meetings to minimize the potential for spillovers. I discuss compliance further in Section 3.3.

similar vein, the first religiously-framed neighbors' rights script presents cleanliness as an act of faith carrying divine reward, while the secular version emphasizes that clean neighborhoods reduce the spread of disease and improve mental wellbeing.

[15] It should be noted, however, that the majority of mosque-goers do not listen to the full sermon and instead arrive only in time for the concluding remarks, so the religious/secular section is the part most likely to be heard by the majority of congregants.

[16] The information sharing sub-treatments append a brief concluding passage in which the imam asks congregants to convey what they have heard to the women in their households, invoking a saying from the Prophet Muhammad to: "Return to your families, stay among them, teach them (the religion), and instruct them to act upon it." In the no-sharing condition, the sermons conclude with no such plea.

### 3.1.2 Outcomes

I pre-registered five primary behavioral outcome measures, three relevant to women's rights and two to neighbors' rights. All primary outcomes involve costly actions for respondents to take rather than self-reported attitudes or behavior. The lab-in-field games are played with real Pakistani rupees, equivalent to 42% of the average daily wage in my setting. For the main analysis, I aggregate the three women's rights outcomes into a standardized women's rights index and the two neighbors' rights outcomes into a standardized neighbors' rights index, following Anderson (2008) and standardizing relative to the control group mean and standard deviation.

The first women's rights outcome is a dictator game in which respondents were given PKR 300 (USD 1.06) and asked how much of it they would like to keep for themselves and how much to donate to a fund that helps women find employment. The amount allocated serves as a costly measure of respondents' willingness to financially support women's economic participation. The second is spouse survey consent: respondents were asked whether they consent to having their wife surveyed independently over the phone, a measure proxying men's willingness to give their wives greater autonomy in a setting where men routinely mediate women's interactions with outsiders. The third outcome is petition signing: respondents were asked whether they were willing to sign a public petition to increase women's representation in Pakistani parliament, as a measure of men's support for women's political participation. For neighbors' rights, the first key outcome is a parallel dictator game in which respondents allocated PKR 300 between themselves and a poor member of their community, serving as a measure of respondents' willingness to take costly actions to support neighbors. The second is community cleanliness: enumerators independently rated the cleanliness of the area where they performed each survey on a 0-10 scale prior to starting the survey, to evaluate whether the imam's plea to clean one's community translated into observable changes in the cleanliness of surrounding communities.

In addition to these costly behavioral measures, I also examine several secondary outcomes capturing self-reported beliefs and behavior to examine the more subtle ways in which sermon content may be having effects over and above my behavioral measures. To capture beliefs, I ask respondents a series of survey questions on a scale from 0-10 and construct indices measuring men's beliefs about women's competence, education, and employment, as well as beliefs about the importance of neighbors, trust in neighbors, and willingness to help neighbors; I examine effects on each of these indices separately as well as on versions pooling all questions. To separate effects on personal beliefs from those on second-order beliefs about social norms or religious beliefs, I modify a subset of these questions to ask what respondents think other men in their community believe or what they think Islamic teachings have to say. To capture self-reported behavior, I ask respondents a series of questions about their time-use, religious practices, as well as various metrics for interactions with neighbors and other household members.

To measure spillover effects within the household, female enumerators survey wives over the phone conditional on both the husband's and wife's consent. Because wives are surveyed over the phone, they cannot be administered the costly behavioral measures, but the self-reported measures follow those for men closely. In addition, I collect measures of wives' well-being and employment behavior, as well as their perceptions of their husbands' time-use and beliefs to capture whether men are indeed behaving differently inside the home. To evaluate impacts on imams, I survey them in person, which allows me to collect both the belief measures and the behavioral outcomes administered to congregants with the exception of spouse survey consent which was not administered to imams in keeping with the cultural norms of my setting.

Finally, I explore equilibrium dynamics at the mosque level by collecting two key outcomes. On the supply side, I measure whether delivering the highly prosocial sermon content causes imams to endogenously change their future sermons - for example, due to changes in imams' own beliefs or due to positive or negative responses from congregants after such content. To this end, I evaluate effects on imams' "non-treatment" sermons - those delivered once the treatments conclude and imams are no longer prompted to deliver any specific content. Enumerators record and transcribe these post-treatment sermons, which are rated on a scale from 0-10 along a number of key content dimensions by a custom GPT developed specifically for the analysis of sermon content transcribed in the local language, yielding a quantitative measure of each mosque's sermon content after the intervention has ended.

On the demand side, I evaluate whether my sermon treatments cause congregation size to change endogenously. Congregation size could increase if prosocial content is what congregants want to hear, decrease if congregants vote with their feet against content they dislike, or remain unchanged if mosque attendance is not very elastic to sermon content. Congregation size is recorded by enumerators who count the number of people present right before the Friday prayer begins, and serves as a revealed-preference measure of demand for the entire congregation, not just the men in my sample.

Impacts on men were assessed 2-4 weeks (an average of 2.6 weeks) after the second and final treatment sermon, while the timing of the wife and imam surveys was more variable, with surveys taking place an average of 5.7 and 3.2 weeks after the conclusion of treatments for wives and imams respectively. Effects on "non-treatment" sermon content and attendance were assessed both in the short and the long run: on the first Friday after the final treatment sermon, and then once more either 6 months or 1 year after the conclusion of treatments.[17] Survey question order was randomized for all respondents to ensure that treatment effects were not confounded by order effects, such as earlier

[17]The staggered treatment roll-out created quasi-random variation in the time between the final treatment and this long run measurement: for 58% of mosques, the long-run measurement took place 6 months after the conclusion of treatments, and for the remaining 42% it took place 1 year after.

questions on one topic priming responses to the other topic or reduced attention at the time of later questions.

### 3.1.3 Estimating Equations

Treatment is assigned at the mosque-level while outcomes are measured at the individual level. I estimate the following primary intent-to-treat (ITT) specification:

$$y_{im} = \alpha + \beta_1 WomenRights_m + \beta_2 NeighborRights_m + \theta_s + \epsilon_{im} \tag{1}$$

where $y_{im}$ is the outcome of interest for individual i in mosque m measured at endline, $WomenRights_m$ and $NeighborRights_m$ are indicators for whether the individual received the women's rights or neighbors' rights treatment sermons respectively, $\theta_s$ represents fixed effects for the randomization strata s, and $\epsilon_{im}$ is the error term.[18] Standard errors are clustered at the level of the treatment, i.e., at the mosque level. The coefficients $\beta_1$ and $\beta_2$ estimate the average ITT effect of more prosocial sermon content on women's and neighbors' rights relative to control group business-as-usual sermons. The same specification is used to estimate effects on wives and imams. For mosque/imam-level outcomes, I estimate analogous specifications at the mosque/imam-level but with robust standard errors. For self-reported outcomes measured at baseline, I include the baseline value of the outcome to improve precision.

In addition to this main specification, to evaluate impacts of secular vs religious framing, I also disaggregate my treatment indicators as follows:

$$\begin{aligned} y_{im} = \alpha &+ \beta_1 WomenReligious_m + \beta_2 WomenSecular_m \\ &+ \beta_3 NeighborReligious_m + \beta_4 NeighborSecular_m + \theta_s + \epsilon_{im} \end{aligned} \tag{2}$$

where $WomenReligious_m$, $WomenSecular_m$, $NeighborReligious_m$, and $NeighborSecular_m$ represent religious and secular framing for the women's rights and neighbors' rights treatments respectively. By comparing whether $\beta_1$ and $\beta_2$ as well as $\beta_3$ and $\beta_4$ are significantly different from each other, I test the extent to which framing of sermon content matters. An analogous specification is used to estimate the effects of my information-sharing sub-treatments.

[18] Strata fixed effects were indicators for: 1) residential/commercial status of the mosque, 2) mosque size attendance tercile at baseline, 3) whether the mosque was a misfit during randomization (for 21 mosques, stratification cell sizes were not divisible by the number of treatment cells, and these misfit mosques were pooled across strata and randomly assigned to treatment groups with the same treatment probabilities used in the main randomization).

## 3.2 Sample and Data

My primary sample consists of male congregants drawn from the universe of 309 Auqaf mosques in Punjab, Pakistan. From each mosque, I randomly sampled and recruited 15 men, with recruitment taking place in two stages. First, the imam introduced the study team during his Friday sermon and asked congregants to volunteer for a research study on the sources of individuals' religious and social beliefs and behavior; an enumerator then collected phone numbers from as many volunteers as possible, with a target of at least 50 per mosque. Then, congregants were called back in random order from the list of phone numbers, and the first 15 to fulfill the recruitment criteria via a brief phone survey were enlisted for the study. The recruitment criteria were: 1) the respondent had to have heard at least 10 minutes of the previous two Fridays' sermons in the same mosque, 2) be aged between 18 and 50, and 3) be married.[19] The first criterion selects regular sermon listeners at the study mosque - excluding those who attend only for the prayer itself, are temporary migrants, or frequently change mosques - and thereby increases the likelihood that respondents were present for the treatment sermons to be delivered several months after recruitment. The second and third recruitment criteria were chosen to increase the likelihood that the men surveyed had relatively young daughters for whom educational and labor market decisions were salient as well as wives to allow for the spouse survey consent outcome and information-sharing sub-treatments to be meaningful.

I recruited a total of 4,667 men at baseline, out of which 3,892 (84%) completed the endline survey. Table A2 reports baseline summary statistics for the endline sample and tests balance across the three main treatment arms. The average respondent is 38 years old with 10 years of education and a monthly household income of approximately PKR 42,000 (USD 150.6). He prays 3.9 of the five daily prayers and attended 3.6 Friday sermons in the past month, consistent with a sample screened on regular mosque attendance. Imam trust is very high at 8.6/10 relative to state trust which is relatively low at 4.7/10. Baseline attitudes toward neighbors are relatively warm: respondents rate the importance of neighbors at 7.3/10 and trust in neighbors at 7.0/10, though willingness to lend to a neighbor is lower at 5.9/10. Attitudes toward women's rights are more conservative: agreement that men make better decisions than women averages 8.1/10, while support for women delaying marriage for education and for the importance of women having jobs are both around 5/10. Table A2 shows that the endline sample is balanced across my main treatment arms on key variables collected at baseline.

Data collection for this project proceeded district by district. Baseline surveys were conducted in person between August 2024 and January 2025. Treatment delivery and endline surveys followed from January 2025 to January 2026 and were staggered in the same district-wise order due to

[19] 8.2% of those called were ineligible because they had not heard the previous two Fridays' sermons in the same mosque, 11.3% were not aged between 18-50, and 11.7% were unmarried. Overall, 71.5% of those administered the phone recruitment survey were found to be eligible for the study.

logistical constraints arising from the geographic dispersion of mosques. The average gap between a respondent's baseline and the treatment sermons at his mosque was 7.7 months. This gap, together with the fact that neither the imams nor the study team ever indicated that the treatment sermons were related to the study or that the study's purpose was to study sermons, reduces the potential for experimenter demand effects that can plague similar interventions in other settings.

## 3.3 First Stage

To evaluate how closely imams followed the treatment scripts, an enumerator was present in the mosque at the time of the treatments to record the audio of the sermons. Recordings were transcribed and the content rated along several dimensions by a custom GPT trained for this purpose. Figure 1 shows average ratings for how closely imams followed the overall message of both sermon scripts. The rating for each mosque is computed by taking the average of the two compliance ratings, one for each of the treatment sermons. As can be seen in the figure, the overall compliance was relatively high with an average rating of 7/10 over all treatment mosques. Importantly, compliance is very similar across the various treatment arms, indicating no differential compliance and allowing for clean comparisons of TEs in the various treatment groups. It should be noted, however, that because imam adherence is imperfect and there is no guarantee that the respondent who attended the mosque at baseline was also present for both treatment sermons in the same mosque, the results in the sections to follow should be read as intent-to-treat effects of attending a treatment mosque at baseline, and likely understate the effect of hearing two highly prosocial sermons in full.[20]

While the compliance ratings tell us how faithfully the imams followed the assigned scripts, they do not tell us how the treatments caused sermon content to change relative to business-as-usual sermons in control mosques. If imams frequently discuss topics such as women's and neighbors' rights in their standard sermons already, then we may be concerned about weak treatments that are unlikely to lead to much change in congregants' beliefs and behavior. In Figure A3, I find that this is not the case by comparing the content of treatment relative to control sermons in treatment weeks. The treatments moved sermon content sharply, causing imams to focus far more on rights (the rights rating in control mosques is 0.8/10 relative to 7.2/10 in treatment mosques) and social topics while somewhat crowding out purely religious topics. The treatments also cause content to become less conservative and less likely to contain hate speech: ratings of conservative content fall from 7.1 to

[20] A key recruitment criteria of the study - that the respondent must have listened to at least 10-minutes of the previous two Fridays' sermon in the same mosque - helps increase the likelihood that the congregant was present at the time of the treatments but does not guarantee it especially since the average time between baseline and treatments was 7.7 months. We could not ask the respondent directly whether they attended the treatment sermon in their original mosque since this would reveal the nature of the treatments thereby increasing the likelihood of surveyor bias but we did ask respondents to report the number of sermons they attended in any mosque in the previous month. The mean for this question was high at 3.6/4, suggesting that most respondents were likely present for both treatment sermons.

5.9, and hate speech, already uncommon at 1.3/10 in control mosques, more than halves to below 0.5/10 in treatment mosques.

This exercise also shows some minor differences between the women's rights and neighbors' rights treatment arms; the neighbors' rights treatments are less likely to focus on historical topics and somewhat less likely to discuss political or economic topics, owing to differences in the treatment scripts.[21] When it comes to the key dimensions of conservativeness and the extent to which rights, religious topics, and social topics are covered, however, both women's and neighbors' rights treatments received very similar ratings.

# 4 Results

In this section, I present the effects of my treatments on men, women, and imams, as well as on mosque-level congregation size and sermon content.

## 4.1 Women's Rights Outcomes

Table 1 presents the effects of the treatment sermons on my primary women's rights outcomes. The outcomes in columns (1)-(3) are allocations in the women's employment dictator game, spouse survey consent rates, and petition signing rates respectively, while column (4) pools these outcomes as a women's rights index. Sermons have large and statistically significant impacts on these costly behavioral outcomes. Men that attended the women's rights mosques allocated PKR 37 (33%) more to a fund supporting women's employment in the dictator game, were 8.5 p.p. (19.5%) more likely to consent to having their wife be surveyed, and 4.9 p.p. (11.2%) more likely to sign a petition supporting women's political participation (though the effect on the petition outcome is statistically insignificant). Pooling these outcomes together, assignment to receive two sermons on women's rights increased the women's rights index by a sizable 0.23 SD.

I next examine effects on self-reported measures of men's beliefs and behavior as they pertain to women. Appendix Table A3 presents treatment effects on belief indices constructed from survey questions about women's competence, education, and employment in columns (1)-(3) respectively, and column (4) pools all questions. Women's rights sermons produce a statistically significant effect of 0.11 SD on the pooled belief index, driven primarily by a large shift in beliefs about women's education (0.18 SD). Effects on beliefs about women's competence and employment are positive (0.07 SD and 0.11 SD respectively) but statistically insignificant. In Table A4, I evaluate effects on

[21] The women's rights scripts had more historical stories to draw from relative to the neighbors' rights sermons given the nature of the topics. Moreover, the women's rights sermons emphasized the importance of women's economic and political participation causing the "political" and "economic" ratings to be relatively high, while the analogous content did not apply for neighbors' rights leading to relatively low ratings.

self-reported behavioral outcomes. Women's rights sermons reduce the amount of time men report their wives spend on household chores and increase men's own comfort in talking to their wives, but have no effect on the time men spend with family members or on men's perceptions of how much time their wives spent working or outside the home.

For comparison, the 0.11 SD and 0.23 SD effects on beliefs and behavior respectively is within the range of other leading gender interventions: Dhar, Jain and Jayachandran (2022) finds that a 2-year school-based gender curriculum for adolescents shifts gender attitudes by 0.18 SD, Field et al. (2021) evaluates the impact of increasing women's financial control and finds impacts of 0.11 SD and 0.09 SD on a labor market engagement index and men's perceived norms respectively, and Bursztyn, González and Yanagizawa-Drott (2020) finds that correcting misperceptions about social norms increases men's willingness to sign their wives up for a job-matching platform by 36%.

## 4.2 Neighbors' Rights Outcomes

Table 2 presents the effects of the treatment sermons on my primary neighbors' rights behavioral outcomes. The outcomes in columns (1) and (2) are allocations in the poor neighbor dictator game and enumerator assessed community cleanliness respectively, while the outcome in column (3) is the neighbors' rights index that pools both outcomes. Men in the neighbors' rights treatment donated PKR 24 (19%) more to help poor neighbors in the dictator game, but I find no statistically significant change in community cleanliness, suggesting that the imam's plea to spend an hour in the following week cleaning the neighborhood did not translate into improvements in community cleanliness observable by enumerators. The overall pooled effect on the neighbors' rights behavioral index is 0.19 SD.

In Appendix Table A5, I look at analogous effects on belief indices about neighbors that quantify the importance individuals place on neighbors, their trust in neighbors, and likelihood of helping neighbors in columns (1)-(3) respectively, while column (4) pools all questions. Examining the results, neighbors' rights sermons produce positive but statistically insignificant effects on all neighbors' rights indices: individuals report a 0.10 SD greater importance of neighbors, a 0.05 SD increase in trust in neighbors, and 0.07 SD greater likelihood of helping neighbors with a pooled index effect of 0.06 SD, suggesting that sermons on women's rights have a larger impact on beliefs relative to sermons on neighbors' rights.

In Appendix Table A6, I look at additional self-reported behavioral outcomes covering men's time use and interactions with neighbors. Neighbors' rights sermons increased the time individuals reported spending with neighbors or in activities that improve their community (19% and 39% respectively), and men also report knowing more neighbors and asking for and receiving help from them more often, though there is no impact on frequency of food sent to neighbors.

Interestingly, while there are no cross-topic effects of the treatments on my behavioral outcomes, Table A5 shows that the women's rights sermons make beliefs about neighbors less prosocial: men who received women's rights sermons report a 0.14 SD lower likelihood of helping neighbors and a significant 0.08 SD reduction in the pooled neighbors' rights index. On the other hand, an analogous cross-domain effect does not occur for sermons on neighbors' rights that actually increased the pooled women's rights index by 0.05 SD as can be seen in Table A3 (though this effect is insignificant). This suggests that the women's rights sermons led to a substitution of prosocial beliefs from one group to another while neighbors' rights sermons improved prosocial beliefs across domains albeit weakly.[22]

## 4.3 Downstream Impacts on Wives

A distinctive feature of the Friday sermon is that it is delivered exclusively to men, so any effects on women must operate through intra-household transmission, by either husbands sharing what they've heard with wives or husbands changing their own behavior in ways that affect their wives, or both. Table 3 presents TEs on the beliefs and well-being of wives that were surveyed by phone conditional on the husband's consent. Wives of men who received women's rights sermons report a 0.19 SD increase in prosocial beliefs about women, and the effect on their own well-being is positive but imprecisely estimated (0.12 SD). On the other hand, wives of men who received sermons on neighbors' rights report no change in prosocial beliefs about neighbors even as they update their beliefs on women's rights, potentially reflecting the downstream impact of the general prosociality effect observed from the neighbors' rights treatments discussed above.

In Table A8, I find no impact on whether wives have a job, are looking for a job, or on beliefs about difficulty of finding a job if wives looked. In Table A9, I investigate wives' perceptions of their husbands behavior and beliefs and find that wives report their husbands spending less time talking to family but perceive them as being more supportive of their daughters' education.

These downstream effects on wives, while necessarily measured conditional on consent and therefore subject to selection concerns, provide suggestive evidence that sermon content permeates throughout households beyond just the direct impact on men. To test for the role of selection in driving these results, in Table A10, I evaluate the extent to which my endline sample of wives was balanced at baseline. Given the low level of overlap between my baseline and endline sample of wives, I can only do this exercise using 19% of my endline observations, but I find that this sub-sample of wives is balanced on almost all variables including the belief indices measured at

[22] In Table A7, I evaluate the effects of my treatments on overall prosociality more explicitly. Columns (1) and (2) reproduce the belief effects on my pooled women's rights and neighbors' rights indices, while column (3) collapses all belief questions into a single prosociality index. The results show that the overall effect of both women's rights and neighbors' rights treatments on overall prosociality is very similar.

baseline.[23] The wives' subsample, however, is small, so I turn to husbands to assess balance more comprehensively. If the wife effects are driven purely by sermons changing the composition of husbands that consent to having their wives surveyed, the endline sample of consenting husbands should be unbalanced at baseline. In Table A10, I find that husbands of the full sample of endline wives are balanced, suggesting that the wife effects are unlikely to be driven purely by selection.

## 4.4 Religious vs Secular Framing

In Table 4, I disaggregate my treatments by whether they were framed in religious or secular language. Doing so reveals a striking asymmetry. For women's rights sermons, religious framing produces substantially larger effects than secular framing: the religiously framed sermons increase the women's rights index by 0.33 SD, while the secularly framed sermons produce a much smaller effect of 0.14 SD (a difference of 0.19 SD, significant at the 1% level). For neighbors' rights sermons, however, the pattern reverses: secularly framed sermons produce a larger and statistically significant effect of 0.25 SD on the neighbors' rights index, while religiously framed sermons produce a smaller effect of 0.14 SD which is insignificant. I am not powered to reject the null of no difference in the case of the neighbors' rights treatment effects, but the difference of 0.11 SD is substantive. Similar prosocial content is thus more effective when grounded in religious arguments for women's rights but in secular arguments for neighbors' rights. I return to this asymmetry in Section 5, where I interpret it through the lens of my theoretical framework.

## 4.5 Supply-Side Effects: Post-Treatment Sermon Content

I next examine whether the changes to sermon content induced by the treatments persisted even once the treatments concluded or if, conversely, imams renege later on what they said during the treatment sermons. In Table 5, I examine the content of the first sermon the imams delivered after the treatments concluded, recorded the first Friday after the final treatment sermon. It is worth reiterating that imams were provided with no instructions for this sermon, so any effect should be interpreted as reflecting the imam's own personal decision as to what to say. Imams who delivered women's rights and neighbors' rights sermons continue to discuss rights, though for women's rights this persistence is concentrated in the religiously framed arm while it is present in both framing arms for neighbors' rights mosques. Sermons in all four arms become significantly less conservative and are less likely to feature hate speech, with the reductions in hate speech statistically significant in both women's rights arms but not in the neighbors' rights arms. In Table A11, I explore effects on additional content dimensions and find that the increase in rights related content comes from less

[23]The exception is whether the wife has a job; wives of husbands in the neighbors' rights treatment group are 13.6 p.p. more likely to have a job at baseline while wives in the women's rights treatment group are balanced.

discussion of social, political, economic, and historical topics, with the specific topics crowded out varying across arms, while there is no change in religious content in any arm.

## 4.6 Demand-Side Responses: Attendance

I conclude the results section by examining whether the sermon treatments generate endogenous impacts on the demand-side by changing the number of people that attend the congregation in treatment mosques. The effects on mosque attendance are presented in Table 6. Column (1) shows that mosques assigned to the religiously framed women's rights sermons experienced a 20% decline in Friday prayer attendance relative to the control group, while the secularly framed women's rights sermons produce a small and statistically insignificant decline of 4%, and no attendance effect is observed for mosques assigned to neighbors' rights sermons. The log specification in column (1) is my preferred specification: attendance is count data and a small number of very large congregations generate outliers that prevent level effects from being detected in the raw headcounts as is shown in column (2). Once the raw data is winsorized, however, the effect re-emerges: winsorizing at the 95th percentile in column (3) yields a statistically significant decline of 65 congregants (18.5% of the control mean) in the religiously framed women's rights arm, while winsorizing at the 90th percentile in column (4) produces a similar decline of 17.1%. These effects are at the mosque-level and so the demand-side attendance effects represent impacts on the full congregation, beyond just those respondents included in the study.

I interpret this as evidence of backlash to the religiously framed women's rights content: while the sermons shifted the beliefs and behavior of some men, it also drove some congregants away. In Table A15, I investigate whether these effects appear after just 1 week of treatment delivery or whether the negative effect of women's rights sermons on attendance take two weeks of prosocial content delivery. Attendance in the religiously framed women's rights mosques after the first treatment sermon falls by almost as much as after the second (a 20% decline relative to control), suggesting that the drop is driven by the initial exposure to women's rights content rather than accumulated with repeated delivery.

## 4.7 Additional Unintended Consequences

The attendance losses documented above are one cost of highly prosocial sermon content, but a policymaker considering sermons as a policy tool may worry about other unintended consequences as well. In Table A13, I explore several such concerns centered on the imams themselves. For instance, we may imagine that utilizing imams for information dissemination emboldens them to speak on topics outside their expertise or that it changes their beliefs about the extent to which religion should be a part of politics and society. The results show that none of the four treatment

arms change imams' beliefs about the extent to which they should discuss non-religious topics (column 1) or give advice on voting or health (columns 3 and 4), and none make them more likely to think that religion should be a part of politics (column 2).[24] We may also worry that being asked to deliver sermons by the state, especially if they contain content which may alienate their audience, may reduce imams' sense of autonomy or satisfaction with their government job. Columns 5 and 6 of Table A13 suggest that this was not the case in any arm; if anything, imams in the secularly framed women's rights arm report marginally higher satisfaction with their autonomy and their job though this effect is only significant at the 10% level.

# 5 Theoretical Framework

In this section, I present a theoretical framework to help with the interpretation of my results, modeling the market for sermons as consisting of both a demand and supply side. On the demand side, congregants solve a two-stage problem each Friday: they first decide whether to attend their mosque and then, having heard the sermon, how to act. On the supply side, imams choose what to say by weighing their own beliefs against what they think their congregations want to hear. I take the model's predictions to the data and show that this standard Bayesian framework rationalizes much of what I find but only partially explains the pattern of treatment effects across framing arms. This motivates an extension in which religious priors are allowed to gate updating from secular signals, which reconciles the remaining patterns. I conclude by turning to other potential mechanisms that may help explain my results.

## 5.1 Demand Side

Mosque-goers have preferences over consumption in this life and the next. Action $y^*$ is God's will on earth and maximizes consumption in the after-life but is uncertain. Similarly, action $z^*$ maximizes consumption during life and is also uncertain. Individuals choose an action to jointly minimize the weighted distance to $y^*$ and $z^*$. I assume quadratic preferences and define $\alpha$ as religiosity, the weight the individual places on conforming to God's will relative to the secular optimum.

Each Friday, individuals solve a two-stage problem: they first decide whether to attend their mosque, and then, having heard the sermon if they attend, choose their action. In the first stage, individual $i$ attends if the consumption value of the sermon, summarized by $b_i$, exceeds the expected

[24] In Table A14, I also evaluate whether male respondents trust their imam more especially with respect to voting or worldly matters and find that this is not the case in any treatment arm. I also do not find that trust in state institutions is lower at endline for men in my treatment groups; if anything, it is higher among men in the secularly framed women's rights arm.

cost of revising one's behavior due to exposure to the sermon:

$$b_i \;\geq\; \kappa_t \, E_i\Big(\Delta_i^2\Big) \tag{3}$$

where $\Delta_i$ is the revision the sermon will induce in the individual's optimal action, the expectation is over the content the individual anticipates hearing at his mosque, and $\kappa_t$ captures the stakes of revising behavior on topic $t$.

In the second stage, individual $i$ chooses $y_i$ to maximize expected utility as follows:

$$-\,E_i\Big(\alpha\,(y_i - y^*)^2 + (1-\alpha)\,(y_i - z^*)^2\Big) \tag{4}$$

The FOC and some algebra give us optimal $y_i$,

$$y_i \;=\; \alpha\, E_i(y^*) + (1-\alpha)\, E_i(z^*) \tag{5}$$

Individuals have priors over $y^*$: $s_i \sim N(y^*, \frac{1}{p_y})$ where $p_y$ indicates the subjective precision reflecting confidence in one's own prior. Similarly, priors over $z^*$ are: $z_i \sim N(z^*, \frac{1}{p_z})$.

The control group gets no signal, so the decision is simple:

$$y_i^C \;=\; \alpha\, s_i + (1-\alpha)\, z_i \tag{6}$$

For the treatment group, attendees get either a religious $R_i \sim N(y^*, \frac{1}{p_R})$ or non-religious $N_i \sim N(z^*, \frac{1}{p_N})$ signal from the imam, depending on whether the sermon is framed in religious or secular terms. As rational Bayesians, treatment individuals update their beliefs and the updated choice is:

$$y_i^{Rel} \;=\; \alpha\,\frac{p_y s_i + p_R R_i}{p_y + p_R} + (1-\alpha)\, z_i \tag{7}$$

$$y_i^{Sec} \;=\; \alpha\, s_i + (1-\alpha)\,\frac{p_z z_i + p_N N_i}{p_z + p_N} \tag{8}$$

Treatment effects are given by subtracting equation (6) from equations (7) and (8). The treatment effect for religious framing is:

$$\Delta_i^{Rel} \;\equiv\; y_i^{Rel} - y_i^C \;=\; \alpha\,\frac{p_R}{p_y + p_R}\,(R_i - s_i) \tag{9}$$

And, the treatment effect for secular framing is:

$$\Delta_i^{Sec} \;\equiv\; y_i^{Sec} - y_i^C \;=\; (1-\alpha)\,\frac{p_N}{p_z + p_N}\,(N_i - z_i) \tag{10}$$

Equations (9) and (10) deliver a few intuitive predictions: treatment effects are increasing in signal surprise, trust in the imam's religious or secular messaging, and, for religious framing, increasing in religiosity which conversely dampens the secular treatment effect. Moreover, since both religious arms are scaled by $\alpha \frac{p_R}{p_y + p_R}$ and both secular arms by $(1-\alpha)\frac{p_N}{p_z + p_N}$, these terms difference out when comparing topics with the same framing. Thus, the ranking of treatment effects across topics must be the inverse of the ranking of the corresponding priors. In a similar vein, condition (3) implies that attendance effects must follow the same ranking as treatment effects, weighted by the stakes of the behavioral revision each topic entails. Finally, the model permits no cross-domain interaction. Religious priors do not enter the secular treatment effect in equation (10), and secular priors do not enter the religious treatment effect in equation (9), so updating from a signal depends only on priors in the domain of that signal.

## 5.2 Supply Side

On the supply side, each Friday each imam $I$ chooses the content of his sermon, $C_I$. He holds his own prior $s_I$ over the optimal action for congregants and derives "preaching" utility from teaching what he believes, but also values the size of his congregation, deriving "pandering" utility from it. By condition (3), congregants exit when a sermon forces a large revision on them, so the attendance the imam expects to lose is increasing in the distance between $C_I$ and his congregation's mean prior, weighted by the mean stakes $\bar{\kappa}$ of the behavior change for congregants. The imam does not observe his congregation's priors directly but infers them from attendance movements. However, since the imam must decide on the content of his sermon before seeing that week's congregation, his estimate of average priors in his congregation, $\bar{s}$, will be based on attendance in prevous weeks. He chooses $C_I$ to maximize

$$-(C_I - s_I)^2 \; - \; \theta\,\bar{\kappa}\;(C_I - \bar{s}_{-1})^2$$

where $\theta$ is the value he places on attendance. The FOC gives

$$C_I^* \;=\; \frac{s_I + \theta\bar{\kappa}\,\bar{s}_{-1}}{1 + \theta\bar{\kappa}} \tag{11}$$

so sermon content is a weighted average of what the imam believes and what he believes his congregation believes.

In the absence of any intervention, content sits close to $\bar{s}_{-1}$, the revision it forces on congregants is small, and condition (3) does not bind: attendance is stable and business-as-usual sermons are conservative. However, delivering the scripts changes imam's own beliefs $s_I$, i.e., self-persuasion. Since $\partial C_I^*/\partial s_I = \frac{1}{1+\theta\bar{\kappa}} > 0$, this will be reflected in the content the imam chooses to deliver in the short run. However in the long run, sustained attendance losses reveal that congregants' priors

sit further from prosocial content than the imam had anticipated, $\bar{s}_{-1}$ is revised downward, and pandering utility pulls content back toward its pre-treatment level even if $s_1$ remains elevated.

## 5.3 Empirics

### 5.3.1 Behavior

Whether the asymmetry in framing effects documented in Table 4 is consistent with the model's predictions depends on the structure of individuals' priors. In Figure 2, I present individuals' baseline religious and secular priors on a 0-10 scale with higher values meaning more prosocial priors. To construct religious priors, I take the average rating individuals gave across a set of questions asking what Islamic teachings say about the importance of, trust in, and obligations toward neighbors in the neighbors' rights case, and about women's decision-making, education, and employment in the women's rights case. Constructing secular priors is less straightforward, so I take the average of two measures: individuals' personal priors and their priors about social norms. Personal priors average ratings from the same set of questions but asking for individuals' own personal beliefs about neighbors and women, while the norms questions ask for individuals' second-order beliefs about people from their community. The two secular measures give almost identical values - suggesting a lack of misperception about social norms - so I take their average as the secular prior, but conclusions are unchanged if I use either measure alone.

As is clear from Figure 2, individuals' priors on women's rights are low in absolute terms, and religious priors are less prosocial than secular priors (3.7/10 and 4.2/10 respectively), while neighbors' rights priors are relatively high, with religious priors exceeding secular priors (8.0/10 vs 6.7/10). Comparing these prior rankings to the treatment effect ranking in Table 4, the data largely conforms to the model's predictions. Within a given topic and across framings, lower priors produce larger effects: religious framing dominates for women's rights (0.33 SD vs 0.14 SD) while secular framing dominates for neighbors' rights (0.25 SD vs 0.14 SD). Even across topics and within framing, the religious treatments follow prior rankings (the far lower religious priors on women's rights produce the largest effect), but comparing the secular arms breaks the pattern. The secular treatment effect for women's rights is very low despite low priors while the secular treatment effect for neighbors is high despite relatively high secular priors. This suggests that updating from secular signals depends on something beyond secular priors, and the data point to religious priors: secular persuasion is weak when religious priors are low and strong exactly when they are high. I formalize this intuition in Section 5.4.

### 5.3.2 Attendance

Turning to attendance, the model predicts backlash where the revision sermons induce is large and the stakes of revising behavior are high. The short-run attendance effects in Table 6 are consistent with this prediction: mosques assigned to the women's rights religiously-framed sermons, where the behavioral change is largest, experienced a 20% decline in Friday prayer attendance relative to the control group, while no attendance effect is observed for mosques assigned to neighbors' rights sermons, whose content was also persuasive but concerns behavior that is likely to be less costly to revise.

The model makes two further predictions that I test here. First, since congregants only observe sermon content by attending, those who leave after the treatment sermons do not observe whether content has since reverted, so the attendance effects should persist. Table 8 presents attendance effects 6 months to 1 year after the conclusion of treatments. I find that the decline in women's rights mosques persists or even grows, consistent with the model. Second, since the revision in equation (9) is increasing in the gap between the sermon's content and congregants' priors, attendance effects should be largest in mosques where congregants' priors are most conservative. In Table A18, I interact my treatment indicators with congregations' average baseline priors constructed using my individual-level data and find that the attendance decline in women's rights mosques is lower in congregations where men are relatively liberal.

### 5.3.3 Sermon Content

Finally, I turn to the supply side. The short-run content effects in Table 5 are consistent with the first prediction of my model: imams who delivered women's rights and neighbors' rights sermons continue to discuss rights, their sermons become less conservative, and are less likely to feature hate speech. The model attributes this persistence to a change in imams' own beliefs, which I test by examining effects on the imams themselves. Table 9 shows that delivering sermons on women's rights causes imams to behave in ways and report beliefs that are more prosocial towards women, with effects concentrated in the religiously framed arm (0.25 SD and 0.20 SD respectively, compared to insignificant effects of 0.14 SD on both outcomes in the secular arm). Interestingly, while delivering sermons on neighbors' rights does not seem to change imams' beliefs and behavior towards neighbors under either framing, it does cause imams to report much more prosocial beliefs about women, mirroring the domain-independent prosociality effect of the neighbors' rights sermon on men and wives. Consistent with the model, the arm in which imams' behavior moves most - women-religious - is also the arm in which content moves most.

This presents an interesting puzzle: imams that delivered sermons on women's rights had their congregations shrink yet continued to deliver more prosocial sermons. The model resolves it through

the timing of imams' learning: imams' own beliefs shift before the attendance signal accumulates, but once it does, the model predicts that content should revert toward its pre-treatment level in the mosques that lost congregants. In Table 7, I examine sermon content 6 months to 1 year after the conclusion of treatments and find that the prosocial shift in women's rights mosques - precisely the mosques that lost congregants - has mostly decayed, while neighbors' rights mosques continue to somewhat discuss rights.

## 5.4 A Model of Cross-Domain Permissibility

While the model explains the attendance and sermon content effects well, as discussed above it still does not fully explain the differential framing effects on men's behavior: treatment effects follow the inverse of prior rankings within topics and across the religious arms, but the secular treatment effect for women's rights is very low despite low priors while the secular treatment effect for neighbors is high despite relatively high secular priors - a reversal that cannot arise under equation (10), since both secular arms share the same scaling and their ranking must follow the inverse ranking of secular priors.

To explain this puzzle, I modify the utility function to allow religious priors to affect the weight individuals place on the secular objective. Individual i now maximizes:

$$-E_i(\alpha(y_i - y^*)^2 + (1-\alpha)\phi(s_i)(y_i - z^*)^2) \tag{12}$$

The distance to the secular optimum is now also weighted by the religious permissibility function $\phi$, which is positively valued and weakly increasing in the religious prior. This allows for the possibility that if individuals' religious priors are too low, they may care less about matching their secular ideals.[25] The first-order condition gives $y_i = \tilde{\alpha}E_i(y^*) + (1-\tilde{\alpha})E_i(z^*)$, where $\tilde{\alpha} = \frac{\alpha}{\alpha+(1-\alpha)\phi(s_i)}$ is the effective religiosity: the weight on the religious objective after accounting for permissibility. Updating proceeds exactly as before and treatment effects become:

[25] In deeply religious contexts, an individual may fully understand and believe a secular signal about, for example, the financial benefits of women working, but feel that acting on it is not permissible given their religious commitments. Conversely, an individual with strong religious priors on the importance of neighbors might feel that their religious foundation licenses them to engage with secular knowledge. In principle, the modification could be symmetric, with secular priors gating religious persuasion through an analogous function $\psi$, but I focus on religious permissibility because the puzzle above relates to the secular treatment effect rankings and also because, in my setting, religion is the domain that determines what is permissible rather than the reverse.

$$
\begin{aligned}
y_i^{Rel} - y_i^C &= \tilde{\alpha} \frac{p_R}{p_y + p_R} (R_i - s_i) \\
&= \frac{\alpha}{\alpha + (1-\alpha)\phi(s_i)} \frac{p_R}{p_y + p_R} (R_i - s_i)
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
y_i^{Sec} - y_i^C &= (1 - \tilde{\alpha}) \frac{p_N}{p_z + p_N} (N_i - z_i) \\
&= \frac{(1-\alpha)\phi(s_i)}{\alpha + (1-\alpha)\phi(s_i)} \frac{p_N}{p_z + p_N} (N_i - z_i)
\end{aligned} \tag{14}
$$

Of course, setting $\phi = 1$ returns the standard model. However, with $\phi \neq 1$, religious priors impact persuasion from secular signals: when religious priors are low, individuals place little weight on the secular objective and under-respond to secular signals, while the reverse holds when religious priors are high. At least descriptively, allowing religious priors to gate secular persuasion in this way reconciles the pattern of treatment effects with the baseline prior results. In the women's rights domain, low religious priors constrain secular persuasion and raise effective religiosity, explaining both the small secular effect and the very large religious effect; in the neighbors' rights domain, high religious priors license engagement with secular arguments while lowering effective religiosity, explaining the reverse.

The updated model's key new prediction, and a sharp departure from standard Bayesian models, is that the secular treatment effect should be increasing in religious priors, which I test in Table A19. I run my religious-secular disaggregated specification but interact the treatment dummies with indicators for above-median baseline religious and secular priors in the relevant domain. For ease of readability of the table, I pool the neighbors' rights group with control in the first column and women's rights group with control in the second, though results are stable in the fully disaggregated specification. For women's rights, the secular treatment effect on the behavioral index nearly triples for respondents with above-median religious priors and this difference is significant at the 10% level. Religious priors thus increase updating from secular signals, in stark contrast to the standard Bayesian case of no cross-domain interaction. For neighbors' rights, the analogous interaction is also positive but imprecisely estimated. All other interactions in the table are either negative or close to 0, consistent with the signal surprise prediction from Bayesian updating.

## 5.5 Other Mechanisms

I next explore additional mechanisms that may be driving treatment effects. The first is information sharing: sermon effects may operate not only through what men hear at the mosque but through

what they subsequently share with others, particularly wives. The direction of any such effect is theoretically ambiguous. Sharing sermon content could amplify effects by prompting useful household conversations, or by raising the threat of cognitive dissonance for men who relay prosocial messages but fail to act on them (Aronson, Fried and Stone, 1992). Alternatively, sharing content designed to increase women's autonomy and well-being could shift household bargaining dynamics in ways men prefer to avoid, dampening effects. My information-sharing sub-treatments, in which imams end the sermon by asking congregants to share what they heard with wives, allow me to test whether exogenously encouraging sharing changes treatment effects.

Table A16 presents treatment effects disaggregated by these sub-treatments. The information sharing plea produces relatively modest and statistically insignificant differences in treatment effects. For women's rights sermons, the sharing treatment yields a coefficient of 0.28 SD compared to 0.18 SD in the no-sharing condition, but I cannot reject the null of equality. For the neighbors' rights sermons, the no-sharing treatment arm actually produces larger effects (0.26 SD vs 0.12 SD), though again the difference is not statistically significant. Taken together, I find no evidence that explicitly encouraging men to share sermon content with wives meaningfully amplifies or attenuates treatment effects, potentially suggesting that the plea adds little to whatever information sharing already occurs naturally after Friday prayers. Notably, the concern that sharing women's rights content could dampen effects by threatening men's household bargaining position finds little support in the data; if anything, the coefficients suggest that sharing content that is prosocial towards women with wives increases rather than decreases treatment effects for women's rights sermons, though this difference is insignificant.

Another mechanism that could be driving treatment effects is social norms: what is unique about sermons as a persuasion tool is that they are delivered publicly to a full congregation, so each listener gets not only a signal about religious doctrine but also about others' beliefs, potentially shifting behavior through effects on second-order beliefs. Alternatively, the social norms component may matter less and effects may run through the effect of sermons on individuals' religious beliefs. In Table A20, I evaluate which channel dominates by decomposing endline beliefs into personal beliefs, beliefs about social norms, and religious beliefs. The results suggest that the impacts of sermons derive from factors beyond norms: the neighbors' rights sermons move religious beliefs (0.17 SD) while leaving perceived norms unchanged, and the women's rights sermons move all three categories by similar magnitudes though the effects are insignificant. Congregants thus appear to view sermons more as a signal about religious truth rather than as information about what others believe, once again highlighting the importance of religious priors for belief updating in this setting.

The norms channel also generates testable predictions across mosques. If sermons persuade by informing listeners about others' beliefs, effects should be larger where that signal is stronger, for example in bigger congregations where more of the community is observed hearing the same

message, and in residential mosques, where congregants know that the men praying beside them are their own neighbors.[26] In Table A21, I interact my pooled treatments with mosque attendance terciles and the residential/commercial status of the mosque, variables that my randomization was stratified by. Treatment effects on women's rights outcomes are very similar across both dimensions. For neighbors' rights, point estimates rise with mosque size and are larger in residential mosques, but no interaction term is statistically distinguishable from zero.

# 6 Conclusion

In this paper, I study how religion drives behavior by focusing on a key religious institution in Muslim societies: the Friday sermon. Working with the Punjab Auqaf Department, I randomized sermon content in 309 state-administered mosques in Punjab, Pakistan, and found sizable effects of just two sermons on women's or neighbors' rights. Men who received the sermons shifted their costly behavior by 0.23 SD and 0.19 SD respectively, with suggestive evidence that effects extended downstream to wives who never heard the sermons themselves. These are large impacts coming from a fairly light-touch intervention: the treatments consisted of roughly an hour of content delivered through an institution congregants already attend, at essentially zero marginal cost to the state. A key limitation of the study is the relatively short time horizon of my individual-level outcomes, measured 2-4 weeks after treatments concluded. While I track mosque-level sermon content and attendance for 6 to 12 months, whether the effects on men's behavior persist over similar horizons remains an open empirical question.

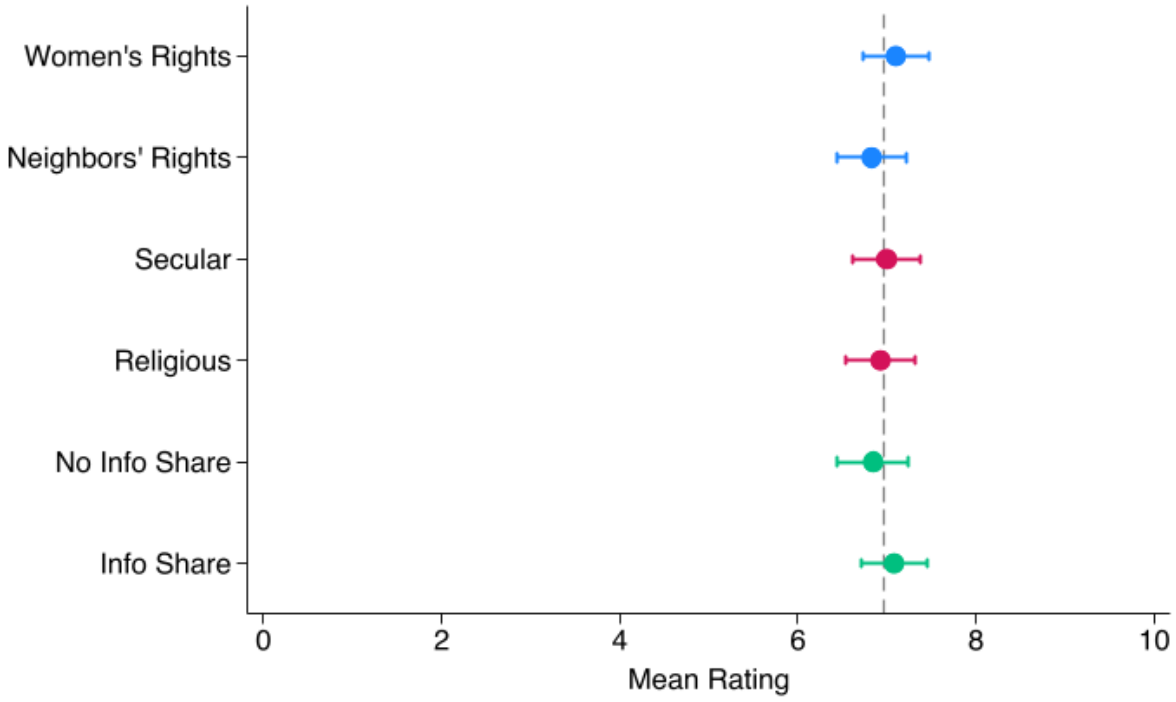
The conclusion that sermons can be systematically used as a policy tool should, however, be made with caution. Mosques that delivered religiously-framed women's rights sermons lost 20% of their congregants, so sustained campaigns of information dissemination through sermons risk alienating the very audiences they seek to reach. The use of sermons for behavioral change also raises important normative concerns. State direction of religious content sits uneasily with longstanding arguments for separating Church from State, and the same machinery that delivered prosocial messages in this study could be used to support more controversial agendas. Indeed, control over sermon content remains actively contested across the Muslim world, with countries ranging from full centralization, as in Turkey and Egypt where the same state-mandated sermon is delivered in all mosques, to regimes like Pakistan's where imams usually have broad discretion over what to say. Yet these concerns stand in tension with the status quo in many Muslim-majority countries, where women's labor force participation remains among the lowest in the world and

[26] Neighbors are precisely the people whose views are most relevant for the behavior of a congregant's wife, who spends her days in the neighborhood, and, in the case of the neighbors' rights sermons, they are also the very people the content concerns.

religious arguments are routinely invoked to justify restrictions on women's education, employment, and mobility. Navigating between these positions is a difficult tightrope for policymakers to walk, and the political economy of interactions between states and religious leaders is an important area for further research.

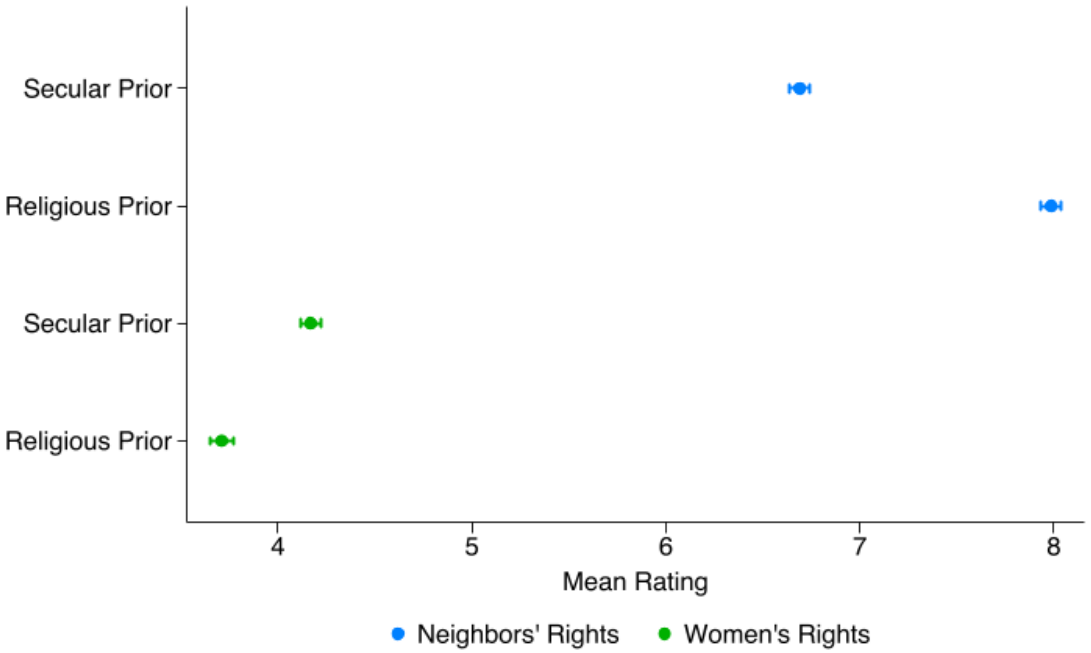
# Tables and Figures

Figure 1: Compliance Across Treatment Arms




Notes: The figure displays mean compliance ratings for the treatment sermons across the six treatment dimensions: topic (Women's Rights vs Neighbors' Rights), framing (Religious vs Secular), and information sharing (Info Share vs No Info Share). Ratings are shown for the 206 treatment mosques only, since control mosques were given no script. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM, which was given both the assigned sermon script and the transcribed sermon and asked: "On a scale from 0 to 10, how closely did the overall message of the imam's sermon match the message of the given script?" The rating for each mosque is the average of the compliance ratings for its two treatment sermons. Dots show the mean rating for each arm and horizontal bars show 95% confidence intervals. The dashed line at 7.0/10 shows the average compliance rating over all treatment mosques. Higher values indicate closer adherence to the assigned script.

Figure 2: Religious vs Secular Priors




Notes: The figure displays mean baseline religious and secular priors of male respondents separately for women's rights and neighbors' rights. Ratings are averages of self-reported survey responses at baseline on a scale from 0-10, with higher values representing more prosocial priors. "Religious Prior" is the average rating across a set of questions asking what Islamic teachings say about the importance of, trust in, and obligations toward neighbors in the neighbors' rights case, and about women's decision-making, education, and employment in the women's rights case. "Secular Prior" is the average of two measures, each constructed by posing the same set of questions but with a different framing: personal priors ask respondents what they themselves believe about the importance of, trust in, and obligations toward neighbors, and about women's decision-making, education, and employment; norms priors ask respondents what they think most people in their community believe about the same set of questions. Dots show the mean rating and horizontal bars show 95% confidence intervals.

Table 1: Behavioral Outcomes: Women's Rights

| Dependent Variable | **Allocation** | **Spouse Survey** | **Petition** | **Pooled** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women's Rights | 37.0819*** | 0.0846** | 0.0494 | 0.2340*** |
| | (7.4395) | (0.0400) | (0.0403) | (0.0512) |
| Neighbors' Rights | 6.2484 | 0.0425 | -0.0230 | 0.0369 |
| | (7.1849) | (0.0429) | (0.0395) | (0.0576) |
| N | 3892 | 3892 | 3892 | 3892 |
| R-Squared | 0.0334 | 0.0054 | 0.0102 | 0.0292 |
| Control Mean | 111.5473 | 0.4326 | 0.4380 | -0.0003 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Column (1) reports the amount in Pakistani rupees allocated to a women's employment fund in a dictator game in which the respondent divides PKR 300 between himself and the fund. Outcome in Column (2) is an indicator equal to one if the respondent consented to having his wife independently surveyed by phone. Column (3) reports an indicator equal to one if the respondent signed a public petition supporting women's political participation in Pakistani parliament. Column (4) pools the three outcomes in Columns (1)-(3) into a Women's Rights Index, an inverse covariance-weighted (ICW) index constructed following Anderson (2008), in standard deviation units standardized relative to the control group mean and standard deviation.

Table 2: Behavioral Outcomes: Neighbors' Rights

| Dependent Variable | **Allocation** | **Enum Clean** | **Pooled** |
|---|---|---|---|
| | (1) | (2) | (3) |
| Women's Rights | 0.9456 | 0.1408 | 0.0559 |
| | (7.2097) | (0.1585) | (0.0660) |
| Neighbors' Rights | 24.0638*** | 0.1873 | 0.1899*** |
| | (7.7036) | (0.1476) | (0.0667) |
| N | 3892 | 3892 | 3892 |
| R-Squared | 0.0285 | 0.0037 | 0.0179 |
| Control Mean | 127.0208 | 7.0485 | 0.0019 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Column (1) reports the amount (PKR) allocated to a poor neighbor in a dictator game in which the respondent divides PKR 300 between himself and a randomly chosen poor person in his community. Column (2) reports enumerator-rated community cleanliness on a 0-10 scale, recorded independently before the survey began. Column (3) pools both outcomes into a Neighbors' Rights Index, an inverse covariance-weighted (ICW) index constructed following Anderson (2008), in standard deviation units standardized relative to the control group mean and standard deviation.

Table 3: Beliefs and Well-Being of Wives

| Dependent Variable | **Women's Rights** | | **Neighbors' Rights** |
|---|---|---|---|
| | **Beliefs** | **Well-Being** | **Beliefs** |
| | (1) | (2) | (3) |
| Women's Rights | 0.1860*** | 0.1232 | 0.0780 |
| | (0.0483) | (0.0967) | (0.0753) |
| Neighbors' Rights | 0.1191** | 0.0763 | -0.0057 |
| | (0.0502) | (0.1109) | (0.0784) |
| N | 618 | 618 | 618 |
| R-Squared | 0.0419 | 0.0157 | 0.0088 |
| Control Mean | -0.0118 | 0.0141 | 0.0091 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are wives of male mosque-goers, surveyed over the phone by female enumerators conditional on consent from both the husband and the wife. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Column (1) is an inverse covariance-weighted (ICW) Women's Rights Beliefs Index following Anderson (2008) pooling wives' self-reported survey responses to all questions quantifying women's competence, education, and employment, in standard deviation units standardized relative to the control group. Column (2) is an ICW Well-Being Index aggregating wives' self-reported responses to questions on happiness, life satisfaction, health, and autonomy, in standard deviation units standardized relative to the control group. Column (3) is an ICW Neighbors' Rights Beliefs Index pooling wives' self-reported survey responses to all questions quantifying the importance of, trust in, and likelihood of helping neighbors, constructed analogously to Column (1).

Table 4: Religious vs Secular Framing

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** |
|---|---|---|
| | (1) | (2) |
| Women Secular | 0.1407** | 0.0279 |
| | (0.0642) | (0.0858) |
| Women Religious | 0.3322*** | 0.0855 |
| | (0.0526) | (0.0761) |
| Neighbors Secular | 0.1189 | 0.2469*** |
| | (0.0735) | (0.0771) |
| Neighbors Religious | -0.0416 | 0.1351 |
| | (0.0677) | (0.0876) |
| N | 3892 | 3892 |
| R-Squared | 0.0417 | 0.0205 |
| Control Mean | -0.0003 | 0.0019 |
| Ho: Women Rel = Women Sec | 0.0020 | 0.5446 |
| Ho: Neighbors Rel = Neighbors Sec | 0.0546 | 0.2550 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Outcome in column (1) is the Women's Rights Index, an inverse covariance-weighted (ICW) index constructed following Anderson (2008) that aggregates: 1) allocation to a women's employment fund in a PKR 300 dictator game, 2) consent to having the respondent's wife independently surveyed by phone, and 3) willingness to sign a petition supporting women's political participation in Pakistani parliament. Outcome in column (2) is the Neighbors' Rights Index which aggregates: 1) allocation to a poor neighbor fund in a PKR 300 dictator game, and 2) enumerator-rated community cleanliness on a 0–10 scale recorded independently before each survey. Treatments are disaggregated into four arms: Women Religious, Women Secular, Neighbors Religious, Neighbors Secular. In the religious framing arms, all content is framed in religious language utilizing Quranic verses, prophetic tradition, and Islamic history; in the secular framing arms the final quarter of each script is framed in secular language, appealing to modern practical, economic, and social concerns. "Ho: Women Rel = Women Sec" reports the p-value from a test of equality of coefficients between the two women's rights framing arms; "Ho: Neighbors Rel = Neighbors Sec" reports the analogous test for the neighbors' rights arms. All index values are in standard deviation units.

Table 5: Post-Treatment Sermon Content

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** | **Conservative Content** | **Hate Speech** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women Secular | 0.0601 | 0.1540 | -0.6656*** | -1.0185*** |
| | (0.2508) | (0.2358) | (0.1470) | (0.2592) |
| Women Religious | 0.9400** | 0.0949 | -0.5667*** | -0.9148*** |
| | (0.3977) | (0.1804) | (0.1393) | (0.2798) |
| Neighbors Secular | -0.2834* | 1.5363*** | -0.5367*** | -0.4332 |
| | (0.1484) | (0.4584) | (0.1559) | (0.3579) |
| Neighbors Religious | -0.0444 | 0.8891** | -0.7073*** | -0.5626 |
| | (0.2133) | (0.3589) | (0.1542) | (0.3420) |
| N | 305 | 305 | 305 | 305 |
| R-Squared | 0.0866 | 0.0973 | 0.1140 | 0.0500 |
| Control Mean | 0.4118 | 0.2157 | 7.1569 | 1.5294 |
| Ho: Women Rel = Women Sec | 0.0407 | 0.8198 | 0.5713 | 0.6265 |
| Ho: Neighbors Rel = Neighbors Sec | 0.1723 | 0.2538 | 0.3777 | 0.7337 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. The outcome in each column is a content rating (0-10 scale) for the first sermon delivered by the imam after the conclusion of treatment sermons. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM. Column (1) rates the extent to which the sermon discussed women's rights; Column (2) the extent to which the sermon discussed neighbors' rights; Column (3) rates the extent to which sermon content was conservative; Column (4) rates the extent to which hate speech was present in the sermon. Higher values indicate greater presence of the relevant content.

Table 6: Post-Treatment Attendance

| Dependent Variable | **Attendance** | | | |
|---|---|---|---|---|
| | **Log(Attendance)** | **Raw** | **Winsor95** | **Winsor90** |
| | (1) | (2) | (3) | (4) |
| Women Secular | -0.0435 | -36.8757 | -3.9741 | 6.6361 |
| | (0.0828) | (23.5111) | (24.1495) | (22.6651) |
| Women Religious | -0.2232** | 26.7550 | -65.1660** | -56.5818** |
| | (0.1029) | (37.3163) | (28.8019) | (26.2370) |
| Neighbors Secular | 0.0454 | 33.0315 | 11.7426 | 18.7582 |
| | (0.0890) | (30.3928) | (27.5342) | (24.3549) |
| Neighbors Religious | 0.0212 | 13.5370 | 10.0373 | 16.4133 |
| | (0.0818) | (27.5961) | (24.1145) | (22.4438) |
| N | 304 | 304 | 304 | 304 |
| R-Squared | 0.6430 | 0.8594 | 0.6685 | 0.6336 |
| Control Mean | 5.6059 | 386.8738 | 352.8447 | 331.2039 |
| Ho: Women Rel = Women Sec | 0.1093 | 0.1017 | 0.0517 | 0.0345 |
| Ho: Neighbors Rel = Neighbors Sec | 0.8031 | 0.5543 | 0.9537 | 0.9317 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the first week's Friday prayer attendance for statistical power. The outcome is Friday prayer headcount measured by enumerators on the first Friday after the conclusion of all treatment sermons. For treatment mosques, this is the Friday following the second treatment sermon; for control mosques, this is two Fridays after the first week of sermon recording. Column (1) reports the natural log of attendance; Column (2) reports the raw headcount; Columns (3) and (4) report headcounts winsorized at the 95th and 90th percentiles, respectively.

Table 7: Post-Treatment Sermon Content (Long Run)

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** | **Conservative Content** | **Hate Speech** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women Secular | 0.2990 | -0.3323*** | 0.0010 | -0.1249 |
| | (0.3764) | (0.1091) | (0.1589) | (0.3515) |
| Women Religious | -0.0315 | -0.0512 | -0.0034 | -0.5989** |
| | (0.3004) | (0.1652) | (0.1515) | (0.2811) |
| Neighbors Secular | 0.1246 | 0.1651 | -0.0531 | -0.3634 |
| | (0.3142) | (0.2354) | (0.1357) | (0.2973) |
| Neighbors Religious | 0.6972** | 0.1585 | 0.0286 | 0.1960 |
| | (0.3315) | (0.2143) | (0.1301) | (0.3391) |
| N | 299 | 299 | 299 | 299 |
| R-Squared | 0.0291 | 0.0243 | 0.0174 | 0.0292 |
| Control Mean | 0.6176 | 0.3529 | 7.8922 | 1.0784 |
| Ho: Women Rel = Women Sec | 0.4388 | 0.0314 | 0.9804 | 0.1746 |
| Ho: Neighbors Rel = Neighbors Sec | 0.1515 | 0.9810 | 0.5547 | 0.1103 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. The outcome in each column is a content rating (0-10 scale) measured by enumerators 6 months to 1 year after the conclusion of all treatment sermons, with control mosques visited over the same period. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM. Column (1) rates the extent to which the sermon discussed women's rights; Column (2) the extent to which the sermon discussed neighbors' rights; Column (3) rates the extent to which sermon content was conservative; Column (4) rates the extent to which hate speech was present in the sermon. Higher values indicate greater presence of the relevant content.

Table 8: Post-Treatment Attendance (Long Run)

| Dependent Variable | **Attendance** | | | |
|---|---|---|---|---|
| | **Log(Attendance)** | **Raw** | **Winsor95** | **Winsor90** |
| | (1) | (2) | (3) | (4) |
| Women Secular | -0.1646 | -88.6834** | -62.0373* | -26.7429 |
| | (0.1040) | (38.8996) | (33.8045) | (28.0781) |
| Women Religious | -0.2676** | -78.7598 | -80.1209** | -61.3555** |
| | (0.1081) | (49.3753) | (38.7095) | (29.0813) |
| Neighbors Secular | -0.0348 | -29.4510 | -20.8517 | 6.6704 |
| | (0.0960) | (40.7863) | (34.3792) | (28.5998) |
| Neighbors Religious | -0.0559 | -15.0755 | -6.6499 | 6.2989 |
| | (0.1020) | (43.1132) | (37.3661) | (29.2748) |
| N | 304 | 304 | 304 | 304 |
| R-Squared | 0.4995 | 0.5089 | 0.4637 | 0.4643 |
| Control Mean | 5.6644 | 411.2621 | 382.5728 | 340.9417 |
| Ho: Women Rel = Women Sec | 0.4111 | 0.8359 | 0.6440 | 0.2876 |
| Ho: Neighbors Rel = Neighbors Sec | 0.8501 | 0.7379 | 0.7082 | 0.9910 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the first week's Friday prayer attendance for statistical power. The outcome is Friday prayer headcount measured by enumerators 6 months to 1 year after the conclusion of all treatment sermons, with control mosques visited over the same period. Column (1) reports the natural log of attendance; Column (2) reports the raw headcount; Columns (3) and (4) report headcounts winsorized at the 95th and 90th percentiles, respectively.

Table 9: Behavior and Beliefs of Imams

| Dependent Variable | **Women's Rights** | | **Neighbors' Rights** | |
|---|---|---|---|---|
| | **Behavior** | **Beliefs** | **Behavior** | **Beliefs** |
| | (1) | (2) | (3) | (4) |
| Women Secular | 0.1423 | 0.1404 | -0.0882 | 0.0583 |
| | (0.1536) | (0.1083) | (0.1407) | (0.1131) |
| Women Religious | 0.2495* | 0.2007** | 0.0686 | -0.0018 |
| | (0.1271) | (0.0967) | (0.1206) | (0.1088) |
| Neighbors Secular | 0.0213 | 0.2168** | -0.0986 | 0.0421 |
| | (0.1282) | (0.1013) | (0.1241) | (0.1022) |
| Neighbors Religious | 0.1551 | 0.1993** | -0.0597 | 0.0185 |
| | (0.1270) | (0.0907) | (0.1376) | (0.1179) |
| N | 244 | 242 | 244 | 243 |
| R-Squared | 0.0295 | 0.0664 | 0.0371 | 0.0065 |
| Control Mean | -0.0000 | -0.0024 | 0.0000 | -0.0101 |
| Ho: Women Rel = Women Sec | 0.5103 | 0.6186 | 0.2456 | 0.6365 |
| Ho: Neighbors Rel = Neighbors Sec | 0.3396 | 0.8734 | 0.7736 | 0.8513 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are imams of Auqaf mosques. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. All outcomes are inverse covariance-weighted (ICW) indices constructed following Anderson (2008), in standard deviation units standardized relative to the control group. Column (1) is the Imams' Women's Rights Behavior Index, an ICW index of: 1) allocation to a women's employment fund in a PKR 300 dictator game, and 2) willingness to sign a petition supporting women's political participation. Column (2) is the Imams' Women's Rights Beliefs Index, constructed from the full set of survey questions quantifying beliefs about the competence, education, and employment of women. Column (3) is the Imams' Neighbors' Rights Behavior Index pooling: 1) allocation to a poor neighbor fund in a PKR 300 dictator game, and 2) enumerator-rated community cleanliness. Column (4) is the Imams' Neighbors' Rights Beliefs Index, constructed from the full set of survey questions quantifying beliefs about the importance of neighbors, trust in neighbors, and likelihood of helping neighbors.

# References


**Acemoglu, Daron, Ali Cheema, Asim I. Khwaja, and James A. Robinson.** 2020. "Trust in State and Nonstate Actors: Evidence from Dispute Resolution in Pakistan." *Journal of Political Economy*, 128(8): 3090–3147.

**Aksoy, Ozan.** 2023. "Preaching to Social Media: Turkey's Friday Khutbas and Their Effects on Twitter." *Socius: Sociological Research for a Dynamic World*, 9.

**Anderson, Michael L.** 2008. "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects." *Journal of the American Statistical Association*, 103(484): 1481–1495.

**Antoun, Richard T.** 1989. *Muslim Preacher in the Modern World: A Jordanian Case Study in Comparative Perspective.* Princeton University Press.

**Aronson, E, Carrie Fried, and J Stone.** 1992. "Overcoming denial and increasing the use of condoms through the induction of hypocrisy." *American journal of public health*, 81: 1636–8.

**Associated Press.** 2016. "Uproar over Egypt's Policy of State-Dictated Friday Sermons." *Associated Press*.

**Avdeenko, Alexandra, Jakob Gärtner, Marc Gillaizeau, Ghida Karbala, Laura Montenbruck, Giulia Montresor, Atika Pasha, and Galina Zudenkova.** 2025. "The Power of Faith: Effects of an Imam-led Information Campaign on Labor Supply and Social Interactions." *Available at SSRN 5017612*.

**Barro, Robert, and Rachel M McCleary.** 2003. "International Determinants of Religiosity." National Bureau of Economic Research Working Paper 10147.

**Becker, Sascha O., and Ludger Woessmann.** 2009. "Was Weber Wrong? A Human Capital Theory of Protestant Economic History." *The Quarterly Journal of Economics*, 124(2): 531–596.

**Borthwick, Bruce M.** 1967. "The Islamic Sermon as a Channel of Political Communication." *Middle East Journal*, 21(3): 299–313.

**Bryan, Gharad, James J Choi, and Dean Karlan.** 2020. "Randomizing Religion: the Impact of Protestant Evangelism on Economic Outcomes*." *The Quarterly Journal of Economics*, 136(1): 293–380.

**Buccione, Giulia.** 2023. "Religious messaging and adaptation to water scarcity: Evidence from Jordan." Working paper.

**Bursztyn, Leonardo, Alessandra L. González, and David Yanagizawa-Drott.** 2020. "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia." *American Economic Review*, 110(10): 2997–3029.

**Campante, Filipe, and David Yanagizawa-Drott.** 2015. "Does Religion Affect Economic Growth and Happiness? Evidence from Ramadan *." *The Quarterly Journal of Economics*, 130(2): 615–658.

**Campbell, BRETT D, MANI Sethuraman, and THOMAS D Steffen.** 2024. "Detecting the Economic Impact of Narratives: Evidence from Religious Sermons." Working Paper, Yale University.

**Clingingsmith, David, Asim Ijaz Khwaja, and Michael Kremer.** 2009. "Estimating the Impact of The Hajj: Religion and Tolerance in Islam's Global Gathering*." *The Quarterly Journal of Economics*, 124(3): 1133–1170.

**Cullen, Claire, Sarthak Joshi, Joseph Vecci, and Julia Talbot-Jones.** 2024. "Female Empowerment and Male Backlash: Experimental Evidence from India." Institute of Labor Economics (IZA) IZA Discussion Paper 17450, Bonn.

**DellaVigna, Stefano, and Ethan Kaplan.** 2007. "The Fox News Effect: Media Bias and Voting*." *The Quarterly Journal of Economics*, 122(3): 1187–1234.

**DellaVigna, Stefano, and Matthew Gentzkow.** 2010. "Persuasion: Empirical Evidence." *Annual Review of Economics*, 2(Volume 2, 2010): 643–669.

**Dhar, Diva, Tarun Jain, and Seema Jayachandran.** 2022. "Reshaping Adolescents' Gender Attitudes: Evidence from a School-Based Experiment in India." *American Economic Review*, 112(3): 899–927.

**Faouzi, Adil.** 2024. "Morocco's Unification of Friday Sermons Sparks Widespread Controversy." *Morocco World News*, Accessed: June 2, 2026.

**Field, Erica, Rohini Pande, Natalia Rigol, Simone Schaner, and Charity Troyer Moore.** 2021. "On Her Own Account: How Strengthening Women's Financial Control Impacts Labor Supply and Gender Norms." *American Economic Review*, 111(7): 2342–75.

**Gaffney, Patrick D.** 1994. *The Prophet's Pulpit: Islamic Preaching in Contemporary Egypt.* Berkeley:University of California Press.

**Haaland, Ingar, Christopher Roth, and Johannes Wohlfart.** 2023. "Designing Information Provision Experiments." *Journal of Economic Literature*, 61(1): 3–40.

**Iannaccone, Laurence R.** 1998. "Introduction to the Economics of Religion." *Journal of Economic Literature*, 36(3): 1465–1495.

**Ibad, Umber Bin.** 2019. *Sufi Shrines and the Pakistani State: The End of Religious Pluralism.* London:I.B. Tauris.

**Iyer, Sriya.** 2016. "The New Economics of Religion." *Journal of Economic Literature*, 54(2): 395–441.

**Jensen, Robert, and Emily Oster.** 2009. "The Power of TV: Cable Television and Women's Status in India*." *The Quarterly Journal of Economics*, 124(3): 1057–1094.

**Kramer, Stephanie, and Dalia Fahmy.** 2018. "The Age Gap in Religion Around the World."

**La Ferrara, Eliana, Alberto Chong, and Suzanne Duryea.** 2012. "Soap Operas and Fertility: Evidence from Brazil." *American Economic Journal: Applied Economics*, 4(4): 1–31.

**La Ferrara, Eliana, and David H Yanagizawa-Drott.** 2026. "Changing Culture and Norms in Developing Countries." National Bureau of Economic Research Working Paper 34784.

**Malik, Jamal.** 1996. *Colonialization of Islam: Dissolution of Traditional Institutions in Pakistan.* New Delhi:Manohar.

**McCleary, Rachel M., and Robert J. Barro.** 2006. "Religion and Economy." *Journal of Economic Perspectives*, 20(2): 49–72.

**Omer, Spahic.** 2013. "The Prophet's Mosque as the Seat of the Prophet's Government." *IslamiCity*.

**Ozdalga, Elisabeth.** 2022. *Pulpit, Mosque and Nation: Turkish Friday Sermons as Text and Ritual.* Edinburgh:Edinburgh University Press.

**Pakistan Bureau of Statistics.** 2023. "7th Population and Housing Census 2023: Provincial Census Report – Punjab." *Islamabad: Government of Pakistan*.

**Pew Research Center.** 2012. "The Global Religious Landscape."

**Prummer, Anja.** 2019. "Religious and Cultural Leaders." *Advances in the Economics of Religion*, , ed. Jean-Paul Carvalho, Sriya Iyer and Jared Rubin, 103–117. Cham:Springer International Publishing.

**Rana, Shahbaz.** 2025. "First Ever Economic Census Shows More Mosques than Factories in Pakistan." *The Express Tribune, August 22*.

**Sarkar, Radha, Ana Sofia Elverdin, and Sebastian Lucek.** 2026. "The Effects of Religious Communication on Gendered Political Attitudes: Experimental Evidence from Colombia." *The Journal of Politics*, 0(ja).

**Schwardmann, Peter, Egon Tripodi, and Joël J. van der Weele.** 2022. "Self-Persuasion: Evidence from Field Experiments at International Debating Competitions." *American Economic Review*, 112(4): 1118–46.

**Shaheed, Farida.** 2010. "Contested Identities: gendered politics, gendered religion in Pakistan." *Third World Quarterly*, 31(6): 851–867. PMID: 20857565.

**Wang, Tianyi.** 2021. "Media, Pulpit, and Populist Persuasion: Evidence from Father Coughlin." *American Economic Review*, 111(9): 3064–92.

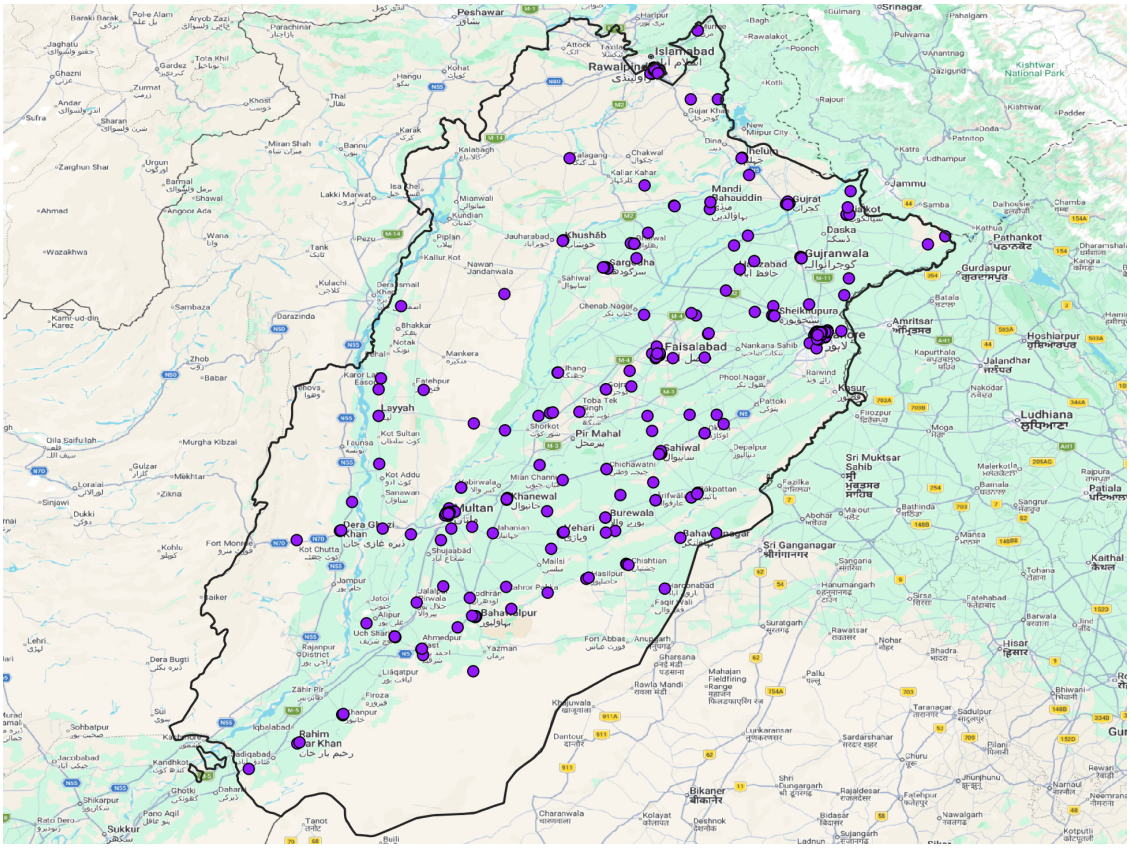
**Weiss, Anita M.** 2014. *Interpreting Islam, Modernity, and Women's Rights in Pakistan.* New York:Palgrave Macmillan.

**World Bank.** 2025. "Labor Force Participation Rate, Female (% of Female Population Ages 15+) (Modeled ILO Estimate)." *World Development Indicators, indicator SL.TLF.CACT.FE.ZS. Source: ILO Modelled Estimates database (ILOEST). Accessed July 8, 2026*.

**Yanagizawa-Drott, David.** 2014. "Propaganda and Conflict: Evidence from the Rwandan Genocide *." *The Quarterly Journal of Economics*, 129(4): 1947–1994.

# Appendix

Figure A1: Spatial distribution of Auqaf mosques




Notes: The map shows the geographic distribution of the 309 Auqaf mosques included in the study across Punjab, Pakistan. The border of Punjab province in Pakistan is shown using the black line while Auqaf mosques are shown as purple dots. Mosques are administered by the Punjab Government's Auqaf and Religious Affairs Department, whose imams are salaried government employees. Study mosques span major urban centers, smaller towns, and rural areas throughout the province. Approximately 13% are located in rural areas, 37% in residential neighborhoods, and 25% have an associated Sufi shrine. Combined Friday prayer attendance across all 309 mosques is approximately 116,000 individuals per week based on baseline counts.

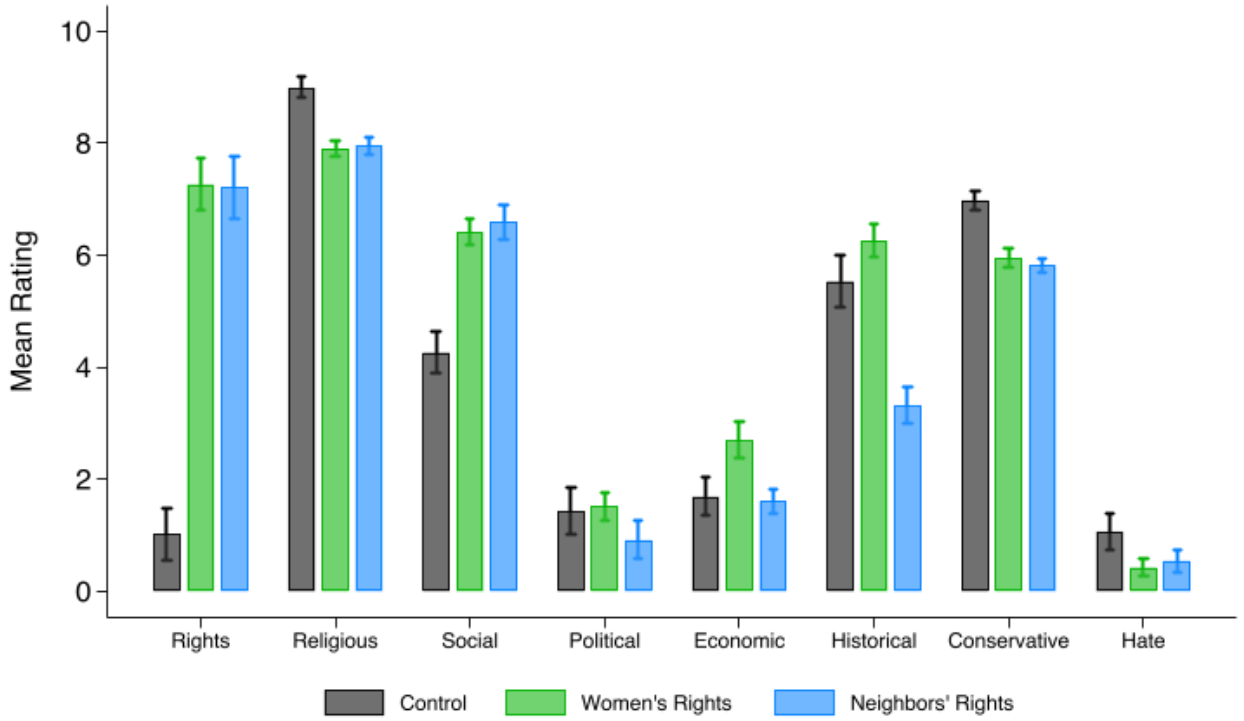
Figure A2: Example Congregation





Notes: Photographs show the setting of a standard Friday sermon at an Auqaf mosque in Punjab, Pakistan. The top image shows the imam delivering a treatment sermon on women's rights before the Friday prayer. The bottom image shows mosque congregants listening to the sermon.

Figure A3: Sermon Content During Treatment Weeks




Notes: The figure displays mean content ratings for sermons delivered during the two treatment weeks separately for control mosques (business-as-usual sermons) and treatment mosques. For treatment mosques, ratings are averages across the two treatment sermon recordings; for control mosques, ratings are from the first week of sermon recording only since only the first and non-treatment weeks were recorded. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM on a scale from 0-10. Eight content dimensions are displayed: rights, religious topics, social topics, political topics, economic topics, historical topics, conservative content, and hate speech. Higher values indicate greater presence of the relevant content.

Table A1: Randomization

| | Religious (Share) | Religious (No Share) | Secular (Share) | Secular (No Share) | Total |
|---|---|---|---|---|---|
| Control | | | | | 103 |
| Women's Rights | 26 | 26 | 25 | 26 | 103 |
| Neighbors' Rights | 26 | 26 | 26 | 25 | 103 |

Notes: The table summarizes the $2 \times 2 \times 2$ factorial randomization design used in this study. Mosques were assigned to one of three topic conditions, Women's Rights, Neighbors' Rights, or Control, with treatment mosques further cross-randomized across two framing conditions (Religious vs Secular) and two information-sharing conditions (Share vs No Share), yielding nine experimental cells. Control mosques were not cross-randomized on framing or sharing sub-treatments since they were given no treatment scripts. Randomization was stratified by mosque Friday prayer attendance tercile and residential/commercial status of the mosque. Cell sizes reflect the number of mosques assigned to each condition out of 309 total.

## Table A2: Endline Sample Balance

| | | Control | | Community Treatment | | Women Treatment | | | |
|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | Mean | SD | Mean | SD | Diff1 | Diff2 |
| **General** | | | | | | | | | |
| Age | 3892 | 37.67 | 7.51 | 37.42 | 7.60 | 37.76 | 7.94 | -0.242 | 0.104 |
| Education | 3883 | 10.00 | 3.73 | 9.61 | 3.81 | 9.86 | 3.64 | -0.381 | -0.121 |
| Income | 3649 | 42039.97 | 24954.99 | 44297.06 | 22549.24 | 42666.11 | 23506.72 | 2,250.294 | 788.294 |
| Daughters Nr | 3892 | 1.31 | 1.18 | 1.34 | 1.23 | 1.33 | 1.21 | 0.028 | 0.015 |
| Pray (Daily) | 3881 | 3.86 | 1.23 | 3.95 | 1.16 | 3.88 | 1.23 | 0.093 | 0.025 |
| Sermons Attend (Monthly) | 3886 | 3.56 | 0.70 | 3.52 | 0.75 | 3.46 | 0.77 | -0.036 | -0.100 |
| Imam Trust | 3886 | 8.57 | 1.80 | 8.47 | 1.64 | 8.53 | 1.64 | -0.100 | -0.036 |
| State Trust | 3861 | 4.71 | 3.09 | 4.72 | 3.02 | 4.48 | 2.89 | 0.007 | -0.225 |
| **Neighbors' Rights** | | | | | | | | | |
| Neighbors Important | 3890 | 7.32 | 2.15 | 7.37 | 2.16 | 7.55 | 2.19 | 0.058 | 0.239 |
| Neighbors Trust | 3885 | 7.00 | 2.13 | 6.93 | 2.24 | 6.94 | 2.22 | -0.069 | -0.055 |
| Neighbor Loan | 3886 | 5.91 | 2.39 | 5.98 | 2.32 | 5.80 | 2.41 | 0.076 | -0.092 |
| Enum Clean | 3892 | 6.89 | 2.16 | 7.08 | 2.00 | 7.07 | 1.97 | 0.187 | 0.183 |
| Belief Index (Neighbors) | 3892 | 0.01 | 0.55 | 0.02 | 0.55 | 0.01 | 0.54 | 0.010 | -0.000 |
| **Women's Rights** | | | | | | | | | |
| Men Better Decisions | 3887 | 8.12 | 2.10 | 7.93 | 2.08 | 7.94 | 2.06 | -0.186 | -0.176 |
| Women Delay Marriage | 3868 | 5.33 | 2.97 | 4.82 | 3.04 | 5.00 | 2.95 | -0.509* | -0.321 |
| Women Job | 3883 | 5.03 | 2.84 | 5.03 | 2.78 | 5.07 | 2.82 | -0.004 | 0.041 |
| Spouse Survey | 3892 | 0.64 | 0.48 | 0.56 | 0.50 | 0.60 | 0.49 | -0.076 | -0.036 |
| Belief Index (Women) | 3892 | -0.01 | 0.43 | 0.03 | 0.44 | 0.03 | 0.45 | 0.032 | 0.040 |

Notes: * 0.10 ** 0.05 *** 0.01. The table reports balance tests for baseline characteristics across treatment arms for the endline men sample. Columns report sample means and standard deviations by treatment arm; Diff1 and Diff2 report the coefficients on the Neighbors' Rights and Women's Rights treatment indicators, respectively, from a regression of each variable on both treatment indicators and stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, with standard errors clustered at the mosque level. The General panel reports demographic and religiosity characteristics: Age (years), Education (years), Income (PKR/month), Daughters Nr (number of daughters), Pray (Daily) (number of the five daily prayers performed, 0–5), Sermons Attend (Monthly) (number of Friday sermons attended in the past four weeks, 0–4), and Imam Trust (0–10 scale, where 0 is "not at all" and 10 is "a lot"). The Neighbors' Rights panel reports Neighbors Important, Neighbors Trust, and Neighbor Loan (0–10 scales measuring, respectively, the importance of neighbors, trust in neighbors, and willingness to loan to a neighbor if asked), Enum Clean (enumerator-rated community cleanliness, 0–10 scale, recorded independently before each survey), and the Belief Index (Neighbors), an inverse covariance-weighted (ICW) index constructed following Anderson (2008) that aggregates all Neighbors' Rights belief items, in standard deviation units standardized relative to the control group mean and standard deviation. The Women's Rights panel reports Men Better Decisions, Women Delay Marriage, and Women Job (0–10 scales measuring agreement with statements about men's decision-making superiority, women delaying marriage for education, and the importance of women having jobs, respectively), Spouse Survey (an indicator equal to one if the respondent's wife completed or agreed to complete the wife survey), and the Belief Index (Women), an ICW index constructed analogously to the Belief Index (Neighbors), aggregating all Women's Rights belief items, also in standard deviation units standardized relative to the control group.

Table A3: Beliefs about Women

| Dependent Variable | Women's Rights | | | |
|---|---|---|---|---|
| | **Competence** | **Education** | **Employment** | **Pooled** |
| | (1) | (2) | (3) | (4) |
| Women's Rights | 0.0660 | 0.1758*** | 0.1135 | 0.1149*** |
| | (0.0534) | (0.0637) | (0.0749) | (0.0438) |
| Neighbors' Rights | -0.0240 | 0.0761 | 0.0356 | 0.0479 |
| | (0.0481) | (0.0615) | (0.0723) | (0.0395) |
| N | 3892 | 3892 | 3892 | 3892 |
| R-Squared | 0.0485 | 0.0919 | 0.0457 | 0.0894 |
| Control Mean | 0.0003 | -0.0009 | 0.0018 | -0.0004 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each index as a control. All outcomes are inverse covariance-weighted (ICW) indices constructed following Anderson (2008), reported in standard deviation units standardized relative to the control group. Column (1) is a Women's Competence Index, aggregating responses to questions that elicit men's beliefs about women's competence. Column (2) is a Women's Education Index, aggregating responses to questions that elicit men's beliefs about women's education. Column (3) is the Women's Employment Index, aggregating responses to questions that elicit men's beliefs about women's employment. Column (4) is a pooled Women's Rights Beliefs Index pooling all questions used to construct the indices in the previous columns into one index. Higher values in all columns imply more prosocial beliefs.

Table A4: Self-Reported Behavior (Women's Rights)

| Dependent Variable | Time-Use (Daily Hrs) | | | Wife | | | |
|---|---|---|---|---|---|---|---|
| | **Wife** | **Daughters** | **Sons** | **Job Hrs** | **Chores Hrs** | **Outside Hrs** | **Comfort** |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Women's Rights | -0.0049 | -0.1439 | 0.0285 | -0.2019 | -0.4135* | -0.0172 | 0.3525** |
| | (0.2351) | (0.2087) | (0.2087) | (0.1824) | (0.2173) | (0.1965) | (0.1729) |
| Neighbors' Rights | 0.0720 | -0.1432 | -0.0611 | -0.2744 | 0.0244 | 0.1585 | 0.1527 |
| | (0.2425) | (0.2109) | (0.2069) | (0.1863) | (0.2103) | (0.1921) | (0.1754) |
| N | 3841 | 2650 | 2822 | 3892 | 3892 | 3892 | 3873 |
| R-Squared | 0.0152 | 0.0222 | 0.0199 | 0.0324 | 0.0272 | 0.0136 | 0.0434 |
| Control Mean | 3.4373 | 2.9026 | 2.8089 | 1.2764 | 5.3048 | 2.6251 | 7.4124 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each outcome as a control where available. Columns (1)-(3) report the respondent's self-reported daily time-use in hours with family members: Column (1) is average daily time spent talking with the respondent's wife; Column (2) is average daily time spent talking with daughters (restricted to respondents who have daughters); Column (3) is average daily time spent talking with sons (restricted to respondents who have sons). Columns (4)-(6) capture the respondent's perceptions of his wife's activities: Column (4) is the respondent's estimate of daily hours his wife spends on paid work; Column (5) is the estimate of daily hours spent on household chores; Column (6) is the estimate of daily hours spent outside the home. Column (8) reports the respondent's self-reported comfort level in talking with his wife, on a 0-10 scale.

Table A5: Beliefs about Neighbors

| Dependent Variable | **Importance** | **Trust** | **Help** | **Pooled** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women's Rights | -0.1149 | -0.0468 | -0.1352** | -0.0867* |
| | (0.0708) | (0.0540) | (0.0593) | (0.0471) |
| Neighbors' Rights | 0.1034 | 0.0546 | 0.0680 | 0.0560 |
| | (0.0723) | (0.0560) | (0.0676) | (0.0575) |
| N | 3892 | 3892 | 3892 | 3892 |
| R-Squared | 0.0702 | 0.0555 | 0.0784 | 0.0691 |
| Control Mean | -0.0019 | 0.0002 | 0.0004 | 0.0006 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each index as a control. All outcomes are inverse covariance-weighted (ICW) indices constructed following Anderson (2008), reported in standard deviation units standardized relative to the control group. Column (1) is the Neighbors' Importance Index, aggregating responses to questions that elicit beliefs about the importance of neighbors. Column (2) is the Neighbors' Trust Index, aggregating responses to questions that elicit beliefs about trust in neighbors. Column (3) is the Neighbors' Help Index, aggregating responses to questions that elicit beliefs about individuals' likelihood of helping neighbors. Column (4) is a pooled Neighbors' Rights Beliefs Index pooling all questions used to construct the indices in the previous columns into one index. Higher values in all columns imply more prosocial beliefs.

Table A6: Self-Reported Behavior (Neighbors' Rights)

| Dependent Variable | **Time-Use (Weekly Hrs)** | | **Neighbors** | | | |
|---|---|---|---|---|---|---|
| | **Neighbors** | **Improving** | **Know** | **Ask Help** | **Receive Help** | **Food Send** |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Women's Rights | -0.0595 | -0.0955 | 1.2855* | -0.2117 | -0.1520 | -0.6373** |
| | (0.1780) | (0.1371) | (0.7389) | (0.2224) | (0.2178) | (0.2847) |
| Neighbors' Rights | 0.4367** | 0.5072*** | 2.1451*** | 0.5289* | 0.6386** | 0.5086 |
| | (0.1942) | (0.1654) | (0.6609) | (0.2724) | (0.2828) | (0.4126) |
| N | 3892 | 3892 | 3892 | 3892 | 3892 | 3892 |
| R-Squared | 0.0726 | 0.0644 | 0.1217 | 0.0742 | 0.0801 | 0.1253 |
| Control Mean | 2.3010 | 1.3218 | 6.7182 | 1.7806 | 2.0608 | 3.5119 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each outcome as a control where available. Columns (1)-(2) report self-reported weekly time-use in hours: Column (1) is weekly time spent interacting with neighbors while Column (2) is weekly time spent on activities improving the community. Columns (3)-(6) report self-reported behavior toward neighbors: Column (3) is the number of neighbors the respondent knows by name; Column (4) is the number of times in the past month the respondent's neighbors asked him for help; Column (5) is the number of times in the past month the respondent received help from his neighbors; Column (6) is the number of times in the past month the respondent's household sent food to neighbors.

Table A7: Prosocial Beliefs

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** | **Overall** |
|---|---|---|---|
| | (1) | (2) | (3) |
| Women's Rights | 0.1149*** | -0.0867* | 0.0318 |
| | (0.0438) | (0.0471) | (0.0301) |
| Neighbors' Rights | 0.0479 | 0.0560 | 0.0497 |
| | (0.0395) | (0.0575) | (0.0304) |
| N | 3892 | 3892 | 3892 |
| R-Squared | 0.0894 | 0.0691 | 0.0735 |
| Control Mean | -0.0004 | 0.0006 | 0.0012 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each index as a control. All outcomes are ICW indices following Anderson (2008) in standard deviation units. Column (1) reproduces the pooled Women's Rights Beliefs Index from Table A3, Column (4). Column (2) reproduces the Neighbors' Rights Beliefs Index from Table A5, Column (4). Column (3) is an Overall Prosocial Beliefs Index that pools all the women's rights and neighbors' rights questions used to construct the indices in columns (1) and (2) into a single overall prosociality index.

Table A8: Employment Behavior of Wives

| Dependent Variable | **Job** | **Job Search** | **Job Search Easy** |
|---|---|---|---|
| | (1) | (2) | (3) |
| Women's Rights | -0.0005 | -0.0220 | -0.4769 |
| | (0.0316) | (0.0257) | (0.3512) |
| Neighbors' Rights | -0.0037 | -0.0360 | -0.2851 |
| | (0.0267) | (0.0239) | (0.3341) |
| N | 618 | 618 | 618 |
| R-Squared | 0.0051 | 0.0202 | 0.0204 |
| Control Mean | 0.0769 | 0.0769 | 4.8791 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are wives of male mosque-goers, surveyed by phone by female enumerators conditional on consent from both the husband and the wife. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Column (1) is an indicator equal to one if the wife has a paid job at endline. Column (2) is an indicator equal to one if the wife is actively searching for a job. Column (3) is a self-reported measure on a scale from 0-10 for how easy it would be to find a job if the wife were to look.

## Table A9: Perceptions of Husbands' Time-Use and Beliefs

| Dependent Variable | **Husbands' Time-Use** | | | **Husbands' Beliefs** | |
|---|---|---|---|---|---|
| | **Wife** | **Sons** | **Daughters** | **Daughter Edu** | **Daughter Job** |
| | (1) | (2) | (3) | (4) | (5) |
| Women's Rights | -1.2358*** | -1.2211*** | -1.0392*** | 0.7659** | -0.2799 |
| | (0.3966) | (0.3960) | (0.3376) | (0.3743) | (0.2898) |
| Neighbors' Rights | -0.6847 | -0.5863 | -0.6322* | 0.1685 | -0.2218 |
| | (0.4148) | (0.4076) | (0.3561) | (0.3812) | (0.3146) |
| N | 618 | 487 | 462 | 618 | 618 |
| R-Squared | 0.0467 | 0.0451 | 0.0425 | 0.0296 | 0.0084 |
| Control Mean | 4.4176 | 3.9653 | 3.7574 | 12.7363 | 5.3901 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are wives of male mosque-goers, surveyed by phone by female enumerators conditional on consent from both the husband and the wife. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Columns (1)–(3) report wives' perceptions of their husbands' daily time-use: (1) hours husband spends talking with the wife; (2) hours spent with sons; (3) hours spent with daughters. Columns (4)–(5) report wives' perceptions of their husbands' beliefs: (4) husband's desired years of schooling for daughters; (5) whether the wife believes her husband would like their daughter to have a job.

## Table A10: Endline Sample Balance (Wives and Husbands)

| | | Control | | Neighbors' Rights | | Women's Rights | | | |
|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | Mean | SD | Mean | SD | Diff1 | Diff2 |
| **Baseline Wives Sample** | | | | | | | | | |
| Age | 113 | 33.88 | 9.08 | 34.69 | 9.32 | 34.82 | 6.20 | 0.812 | 0.937 |
| Education | 113 | 8.61 | 4.62 | 8.57 | 4.90 | 7.29 | 5.00 | -0.035 | -1.317 |
| Job | 113 | 0.00 | 0.00 | 0.10 | 0.30 | 0.05 | 0.23 | 0.095 | 0.053 |
| Job Search | 113 | 0.06 | 0.24 | 0.05 | 0.22 | 0.00 | 0.00 | -0.013 | -0.061 |
| Children Nr | 113 | 3.15 | 1.79 | 3.36 | 2.14 | 3.00 | 1.59 | 0.206 | -0.152 |
| Daily Pray | 113 | 4.09 | 1.28 | 4.07 | 1.35 | 4.08 | 1.10 | -0.019 | -0.012 |
| Well-Being | 113 | 0.00 | 0.83 | 0.10 | 0.65 | -0.04 | 0.75 | 0.098 | -0.050 |
| Beliefs Women | 113 | 0.04 | 0.50 | 0.05 | 0.50 | 0.09 | 0.32 | 0.009 | 0.051 |
| Beliefs Neighbors | 113 | -0.01 | 0.56 | 0.04 | 0.54 | -0.04 | 0.44 | 0.048 | -0.032 |
| **Full Wives Sample** | | | | | | | | | |
| Age (Husband) | 618 | 37.97 | 7.14 | 37.07 | 7.32 | 37.65 | 7.76 | -0.904 | -0.321 |
| Education (Husband) | 618 | 10.08 | 3.60 | 9.33 | 3.54 | 10.01 | 3.54 | -0.757 | -0.069 |
| Income (Husband) | 585 | 42807.02 | 25000.31 | 42000.00 | 23371.35 | 40931.37 | 21624.09 | -807.018 | -1,875.645 |
| Children Nr (Husband) | 616 | 2.82 | 1.63 | 2.80 | 1.74 | 2.57 | 1.66 | -0.021 | -0.249 |
| Daily Pray (Husband) | 613 | 3.81 | 1.25 | 3.92 | 1.20 | 3.83 | 1.31 | 0.112 | 0.022 |
| Beliefs Women (Husband) | 618 | 0.03 | 0.40 | 0.03 | 0.45 | 0.05 | 0.42 | -0.002 | 0.014 |
| Beliefs Neighbors (Husband) | 618 | 0.01 | 0.56 | 0.07 | 0.57 | -0.01 | 0.51 | 0.062 | -0.017 |
| Spouse Survey (Husband) | 618 | 0.70 | 0.46 | 0.67 | 0.47 | 0.58 | 0.50 | -0.029 | -0.125 |
| Enum Clean (Husband) | 618 | 6.77 | 2.12 | 6.96 | 1.93 | 6.83 | 2.19 | 0.184 | 0.051 |

Notes: * 0.10 ** 0.05 *** 0.01. The table reports balance tests for pre-treatment baseline characteristics across treatment arms for the wives' sample. Columns report sample means by treatment arm (Control, Neighbors' Rights, and Women's Rights) and the p-value from an F-test of joint equality across arms, from a regression of each characteristic on treatment indicators and stratification fixed effects with standard errors clustered at the mosque level. The upper panel tests balance for wives in the endline sample that could be matched to a baseline wife observation (approximately 19% of endline wives). The lower panel tests balance of husband characteristics for the full endline sample of wives.

Table A11: Post-Treatment Sermon Content

| Dependent Variable | **Religious** | **Social** | **Political** | **Economic** | **Historical** |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Women Secular | 0.1479 | -1.0420*** | -0.8041* | -0.5004 | -0.8176* |
| | (0.2410) | (0.3379) | (0.4100) | (0.3279) | (0.4243) |
| Women Religious | 0.0135 | 0.0968 | -0.4897 | -0.2875 | -0.3853 |
| | (0.2148) | (0.3561) | (0.4411) | (0.3487) | (0.3456) |
| Neighbors Secular | -0.0569 | -0.2799 | -0.2985 | -0.3208 | -1.3577*** |
| | (0.2027) | (0.3615) | (0.4689) | (0.2977) | (0.4166) |
| Neighbors Religious | 0.0525 | -0.6719* | -0.5218 | -0.6853** | -0.8907** |
| | (0.2043) | (0.3826) | (0.4802) | (0.2804) | (0.4103) |
| N | 305 | 305 | 305 | 305 | 305 |
| R-Squared | 0.0130 | 0.0501 | 0.0168 | 0.0314 | 0.0508 |
| Control Mean | 8.7549 | 4.6176 | 2.2157 | 1.9608 | 5.7059 |
| Ho: Women Rel = Women Sec | 0.6095 | 0.0038 | 0.4913 | 0.5979 | 0.3634 |
| Ho: Neighbors Rel = Neighbors Sec | 0.6210 | 0.3655 | 0.6821 | 0.2513 | 0.3685 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. The outcome in each column is a content rating (0-10 scale) for the first sermon delivered by the imam after the conclusion of treatment sermons. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM. Column (1)-(5) rate the extent to which the sermons discussed religious, social, political, economic, and historical topics respectively. Higher values indicate greater presence of the relevant content.

Table A12: Post-Treatment Sermon Content (Long Run)

| Dependent Variable | **Religious** | **Social** | **Political** | **Economic** | **Historical** |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Women Secular | 0.1239 | -0.6882* | -0.0291 | -0.6997* | 0.5316 |
| | (0.1113) | (0.3990) | (0.3498) | (0.3885) | (0.4538) |
| Women Religious | 0.1924** | -0.4937 | -0.5847** | -0.6004 | -0.2846 |
| | (0.0970) | (0.4449) | (0.2859) | (0.4088) | (0.4258) |
| Neighbors Secular | 0.1668* | -0.4522 | -0.5923** | -0.2565 | 0.4881 |
| | (0.0946) | (0.4295) | (0.2657) | (0.4370) | (0.4579) |
| Neighbors Religious | -0.1703 | -0.2551 | -0.1468 | -0.4294 | -0.0341 |
| | (0.2212) | (0.4224) | (0.3292) | (0.4092) | (0.5010) |
| N | 299 | 299 | 299 | 299 | 299 |
| R-Squared | 0.0399 | 0.0295 | 0.0404 | 0.0428 | 0.0268 |
| Control Mean | 9.8137 | 4.7745 | 1.2255 | 2.7059 | 5.8824 |
| Ho: Women Rel = Women Sec | 0.2820 | 0.6974 | 0.1253 | 0.8101 | 0.0923 |
| Ho: Neighbors Rel = Neighbors Sec | 0.0999 | 0.6962 | 0.1701 | 0.7070 | 0.3475 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. The outcome in each column is a content rating (0-10 scale) for a sermon delivered by the imam 6 months to 1 year after the conclusion of treatment sermons, recorded during a return visit to each mosque. Sermon audio was recorded by enumerators, transcribed, and then rated by a custom LLM. Column (1)-(5) rate the extent to which the sermons discussed religious, social, political, economic, and historical topics respectively. Higher values indicate greater presence of the relevant content.

Table A13: Domain Expansion and Job Satisfaction of Imams

| Dependent Variable | Domain Expansion | | | | Job Satisfaction | |
|---|---|---|---|---|---|---|
| | **Non-Rel** | **Rel Politics** | **Vote Advice** | **Health Politics** | **Autonomy** | **Satisfaction** |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Women Secular | 0.0632 | 0.0396 | -0.4168 | -0.0672 | 0.6760* | 0.4663* |
| | (0.4316) | (0.4072) | (0.6212) | (0.4030) | (0.3633) | (0.2693) |
| Women Religious | -0.5943 | -0.3306 | -0.6422 | 0.0988 | 0.2466 | 0.1078 |
| | (0.4464) | (0.4486) | (0.6183) | (0.3986) | (0.4103) | (0.2957) |
| Neighbors Secular | 0.1497 | -0.4576 | -1.0383* | -0.4370 | 0.3464 | -0.0421 |
| | (0.4070) | (0.4456) | (0.5380) | (0.4099) | (0.3730) | (0.3252) |
| Neighbors Religious | -0.0388 | -0.0966 | -0.3906 | -0.5991 | -0.0392 | -0.1609 |
| | (0.3789) | (0.4134) | (0.5390) | (0.4693) | (0.3430) | (0.2797) |
| N | 241 | 242 | 242 | 242 | 238 | 236 |
| R-Squared | 0.0169 | 0.0213 | 0.0276 | 0.0167 | 0.0223 | 0.0423 |
| Control Mean | 7.4865 | 7.8667 | 5.5333 | 7.2533 | 7.7838 | 8.4306 |
| Ho: Women Rel = Women Sec | 0.2292 | 0.4850 | 0.7597 | 0.7310 | 0.3366 | 0.2234 |
| Ho: Neighbors Rel = Neighbors Sec | 0.6842 | 0.4961 | 0.2809 | 0.7676 | 0.3296 | 0.7208 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are imams of Auqaf mosques. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. All outcomes are on a 0–10 scale. Column (1) reports whether imams think they should have more say in non-religious matters. Column (2) reports the extent to which imams think religion should play a bigger role in politics. Column (3) reports whether imams think they should be advising congregants on how to vote. Column (4) reports whether imams think they should be advising congregants on health matters. Column (5) reports self-reported satisfaction with the autonomy the imam says they have in their government job. Column (6) reports imams' overall self-reported satisfaction with their government job.

Table A14: Trust

| Dependent Variable | Imam Trust | | | | State |
|---|---|---|---|---|---|
| | **Overall** | **Voting** | **Religious** | **Worldly** | **Overall** |
| | (1) | (2) | (3) | (4) | (5) |
| Women Secular | 0.2782 | -0.0349 | 0.2040 | -0.0384 | 0.5826** |
| | (0.2214) | (0.4145) | (0.1854) | (0.3835) | (0.2481) |
| Women Religious | 0.0352 | 0.1121 | -0.0818 | 0.1493 | 0.0276 |
| | (0.2374) | (0.4243) | (0.1933) | (0.3333) | (0.2447) |
| Neighbors Secular | 0.0514 | 0.2671 | 0.0138 | 0.2855 | 0.1035 |
| | (0.2599) | (0.3843) | (0.2073) | (0.3271) | (0.2456) |
| Neighbors Religious | 0.2756 | 0.1004 | 0.0951 | 0.1503 | 0.1061 |
| | (0.2421) | (0.3755) | (0.1935) | (0.3324) | (0.2777) |
| N | 3882 | 3854 | 3885 | 3874 | 3837 |
| R-Squared | 0.0391 | 0.0570 | 0.0350 | 0.0351 | 0.0925 |
| Control Mean | 7.8128 | 5.3624 | 8.1278 | 6.3840 | 5.3841 |
| Ho: Women Rel = Women Sec | 0.3489 | 0.7699 | 0.2036 | 0.6666 | 0.0617 |
| Ho: Neighbors Rel = Neighbors Sec | 0.4491 | 0.7007 | 0.7363 | 0.7245 | 0.9936 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each outcome as a control. All outcomes are on a 0–10 scale. Column (1) is overall trust in the respondent's imam. Column (2) is trust in the imam for guidance on voting decisions. Column (3) is trust in the imam on religious matters. Column (4) is trust in the imam on worldly (non-religious) matters. Column (5) is respondents' overall trust in state institutions.

Table A15: Post-Treatment Attendance (Week 2)

| Dependent Variable | **Log(Attendance)** | **Raw** | **Winsor99** | **Winsor95** |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women Secular | -0.0501 | -41.4071** | -39.2622* | -10.0528 |
| | (0.0753) | (20.0444) | (21.0897) | (24.0324) |
| Women Religious | -0.2181* | 14.3745 | -47.1840* | -57.0017* |
| | (0.1144) | (26.4917) | (25.2824) | (30.1634) |
| Neighbors Secular | 0.0381 | 12.5475 | -5.7466 | 6.9440 |
| | (0.0909) | (30.1507) | (28.3387) | (28.8563) |
| Neighbors Religious | -0.0798 | 18.5928 | -8.8099 | -4.4398 |
| | (0.1152) | (47.3944) | (34.4335) | (29.4922) |
| N | 303 | 303 | 303 | 303 |
| R-Squared | 0.5736 | 0.8271 | 0.7958 | 0.6273 |
| Control Mean | 5.6059 | 386.8738 | 386.8738 | 352.6117 |
| Ho: Women Rel = Women Sec | 0.1532 | 0.0294 | 0.7473 | 0.1482 |
| Ho: Neighbors Rel = Neighbors Sec | 0.3522 | 0.9033 | 0.9357 | 0.7433 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the first week's Friday prayer attendance for statistical power. The outcome is Friday prayer headcount measured by enumerators on the day of the second treatment sermon, while for control mosques, it is headcount two Fridays after the first week of sermon recording. Column (1) reports the natural log of attendance; Column (2) reports the raw headcount; Columns (3) and (4) report headcounts winsorized at the 99th and 95th percentiles, respectively.

Table A16: Information Sharing

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** |
|---|---|---|
| | (1) | (2) |
| Women No Share | 0.1848*** | 0.0265 |
| | (0.0613) | (0.0741) |
| Women Share | 0.2825*** | 0.0849 |
| | (0.0591) | (0.0881) |
| Neighbors No Share | 0.0907 | 0.2636*** |
| | (0.0783) | (0.0832) |
| Neighbors Share | -0.0172 | 0.1155 |
| | (0.0622) | (0.0809) |
| N | 3892 | 3892 |
| R-Squared | 0.0334 | 0.0221 |
| Control Mean | -0.0003 | 0.0019 |
| Ho: Women Share = Women No Share | 0.1301 | 0.5426 |
| Ho: Neighbors Share = Neighbors No Share | 0.1951 | 0.1282 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator. Outcome in column (1) is the Women's Rights Index, an inverse covariance-weighted (ICW) index constructed following Anderson (2008) that aggregates: 1) allocation to a women's employment fund in a PKR 300 dictator game, 2) consent to having the respondent's wife independently surveyed by phone, and 3) willingness to sign a petition supporting women's political participation in Pakistani parliament. Outcome in column (2) is the Neighbors' Rights Index which aggregates: 1) allocation to a poor neighbor fund in a PKR 300 dictator game, and 2) enumerator-rated community cleanliness on a 0–10 scale recorded independently before each survey. Treatments are disaggregated into four arms: Women Share, Women No Share, Neighbors Share, Neighbors No Share. In the sharing arm, sermons conclude with an explicit plea from the imam asking congregants to share what they have heard in the sermon with wives, while in the no sharing arm, sermons do not conclude with such a plea. "Ho: Women Share = Women No Share" reports the p-value from a test of equality of coefficients between the women's rights sharing/no sharing arms; "Ho: Neighbors Share = Neighbors No Share" reports the analogous test for the neighbors' rights arms. All index values are in standard deviation units.

Table A17: Post-Treatment Attendance by Tercile of Distance to Nearest Mosque

| Dependent Variable | **Log(Attendance)** | |
|---|---|---|
| | **Short Run** | **Long Run** |
| | (1) | (2) |
| Women Secular | -0.2787** | -0.1755 |
| | (0.1295) | (0.2231) |
| Women Religious | -0.3083 | -0.3656* |
| | (0.2069) | (0.1992) |
| Women Secular | 0.3248* | 0.0753 |
| × Distance T2 | (0.1780) | (0.2598) |
| Women Religious | 0.2220 | 0.1661 |
| × Distance T2 | (0.2412) | (0.2555) |
| Women Secular | 0.3179* | 0.0574 |
| × Distance T3 | (0.1873) | (0.2640) |
| Women Religious | -0.0706 | 0.0993 |
| × Distance T3 | (0.2776) | (0.2694) |
| N | 293 | 293 |
| R-Squared | 0.6586 | 0.5003 |
| Control Mean | 5.5853 | 5.6133 |

Notes: * 0.10 ** 0.05 *** 0.01. Unit of observation is the mosque. Standard errors are robust. This table adds an interaction between each treatment arm and the tercile of distance from the study mosque to the next nearest mosque. The comparison group pools the neighbors' rights mosques with the pure control mosques.

Table A18: Behavior and Attendance by Baseline Priors

| Dependent Variable | **Behavioral Index** | | **Log(Attendance)** | |
|---|---|---|---|---|
| | **Women's Rights** | **Neighbors' Rights** | **Short Run** | **Long Run** |
| | (1) | (2) | (3) | (4) |
| Women Secular | 0.1432** | 0.0255 | -0.0313 | -0.1782* |
| | (0.0653) | (0.0849) | (0.0802) | (0.1029) |
| Women Religious | 0.3316*** | 0.0874 | -0.2199** | -0.2651** |
| | (0.0522) | (0.0756) | (0.1024) | (0.1082) |
| Neighbors Secular | 0.1215 | 0.2471*** | 0.0426 | -0.0174 |
| | (0.0739) | (0.0733) | (0.0874) | (0.0937) |
| Neighbors Religious | -0.0434 | 0.1381 | 0.0251 | -0.0592 |
| | (0.0671) | (0.0862) | (0.0834) | (0.1037) |
| Women Secular | -0.1793 | -0.0038 | -0.2434 | 0.3203 |
| × Prior | (0.1353) | (0.1023) | (0.2427) | (0.2586) |
| Women Religious | -0.0707 | -0.0446 | 0.1511 | 0.2698 |
| × Prior | (0.0941) | (0.1140) | (0.3409) | (0.2899) |
| Neighbors Secular | -0.1733 | 0.1246 | 0.0657 | 0.4046* |
| × Prior | (0.1164) | (0.0995) | (0.2090) | (0.2367) |
| Neighbors Religious | -0.1576 | -0.1928 | -0.1102 | -0.1536 |
| × Prior | (0.1172) | (0.1238) | (0.1847) | (0.2455) |
| N | 3892 | 3892 | 304 | 304 |
| R-Squared | 0.0449 | 0.0317 | 0.6455 | 0.5079 |
| Control Mean | -0.0003 | 0.0019 | 5.6059 | 5.6644 |

Notes: * 0.10 ** 0.05 *** 0.01. Columns (1) and (2) are individual-level while columns (3) and (4) are mosque-level, collapsing the priors to the congregation level. The priors in columns 1 and 2 are over women's rights and neighbors' rights respectively. The priors in columns 3 and 4 are priors over women's rights in the first two interaction rows and priors over neighbors' rights in the second two interaction rows.

Table A19: Heterogeneity by Baseline Religious and Secular Priors

| Dependent Variable | **Women's Rights** | **Neighbors' Rights** |
|---|---|---|
| | (1) | (2) |
| Religious | 0.3000*** | 0.3605*** |
| | (0.0681) | (0.1028) |
| Secular | 0.0957 | 0.2808*** |
| | (0.0974) | (0.0967) |
| Religous x | 0.0298 | -0.2270** |
| 1(Religious Prior > Median) | (0.0756) | (0.1020) |
| Secular x | 0.1874* | 0.0939 |
| 1(Religious Prior > Median) | (0.0981) | (0.1104) |
| Religious x | -0.0175 | -0.2332** |
| 1(Secular Prior > Median) | (0.0709) | (0.1009) |
| Secular x 1(Secular | -0.1237 | -0.2152** |
| Prior > Median) | (0.1023) | (0.1003) |
| N | 3648 | 3759 |
| R-Squared | 0.0406 | 0.0463 |
| Control Mean | -0.0003 | 0.0019 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers from 309 mosques. Standard errors clustered at mosque-level. Outcome in column 1 is index that collapses my primary women's rights outcomes: 1) allocation in dictator game, 2) spouse survey consent, 3) petition consent. Outcome in column 2 is index that collapses my primary neighbors' rights outcomes: 1) allocation in dictator game, 2) enumerator-measured community cleanliness. Indices constructed using Anderson (2008) and units are standard deviations. "Religious Prior" represents average religious beliefs expressed at baseline. "Secular Prior" represents average secular beliefs (average of beliefs about social norms and personal beliefs) expressed at baseline.

Table A20: Personal vs Norms vs Religious Beliefs

| Dependent Variable | **Women's Rights** | | | **Neighbors' Rights** | | |
|---|---|---|---|---|---|---|
| | **Personal** | **Norms** | **Religious** | **Personal** | **Norms** | **Religious** |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Women's Rights | 0.0711 | 0.0905 | 0.0974 | -0.1174* | -0.1528** | 0.0441 |
| | (0.0688) | (0.0670) | (0.0845) | (0.0665) | (0.0731) | (0.0938) |
| Neighbors' Rights | -0.0506 | -0.0642 | -0.0281 | 0.1067 | 0.0189 | 0.1734* |
| | (0.0608) | (0.0646) | (0.0799) | (0.0699) | (0.0788) | (0.0957) |
| N | 3892 | 3870 | 3886 | 3892 | 3871 | 3889 |
| R-Squared | 0.0606 | 0.0312 | 0.0500 | 0.0838 | 0.0807 | 0.0525 |
| Control Mean | 0.0015 | -0.0006 | -0.0014 | -0.0032 | -0.0045 | -0.0036 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include stratification fixed effects for mosque attendance tercile, residential/commercial status, and a misfit indicator, as well as the baseline value of each index as a control. All outcomes are ICW indices following Anderson (2008) in standard deviation units. For women's rights (Columns 1–3): 1) Personal Beliefs Index, constructed from the respondent's own survey responses on women's competence, education, and employment; 2) Social Norms Index, constructed from analogous questions about what other men in the community think; 3) Religious Prior Index, constructed from analogous questions about what Islamic teaching says on women's rights. For neighbors' rights (Columns 4–6): 4) Personal Beliefs Index for neighbors; 5) Social Norms Index for neighbors; 6) Religious Prior Index for neighbors.

Table A21: Heterogeneity on Mosque-Level Dimensions

| Dependent Variable | **Women's Rights** | | **Neighbors' Rights** | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Women | 0.2408*** | 0.2357 | 0.1368 | -0.1351 |
| | (0.0891) | (0.1527) | (0.1076) | (0.1993) |
| Neighbors | 0.0989 | 0.2616 | 0.1094 | 0.0800 |
| | (0.0939) | (0.1743) | (0.1120) | (0.1999) |
| WomenxAttendance2 | -0.0077 | | -0.0663 | |
| | (0.1173) | | (0.1497) | |
| WomenxAttendance3 | -0.0126 | | -0.1797 | |
| | (0.1332) | | (0.1652) | |
| NeighborsxAttendance2 | -0.1594 | | 0.0857 | |
| | (0.1358) | | (0.1655) | |
| NeighborsxAttendance2 | -0.0214 | | 0.1538 | |
| | (0.1415) | | (0.1582) | |
| WomenxResidential | | -0.0026 | | 0.1392 |
| | | (0.1099) | | (0.1328) |
| NeighborsxResidential | | -0.1646 | | 0.0809 |
| | | (0.1168) | | (0.1426) |
| N | 3892 | 3892 | 3892 | 3892 |
| R-Squared | 0.0319 | 0.0325 | 0.0239 | 0.0193 |
| Control Mean | -0.0003 | -0.0003 | 0.0019 | 0.0019 |

Notes: * 0.10 ** 0.05 *** 0.01. Respondents are male mosque-goers recruited from 309 Auqaf mosques in Punjab, Pakistan. Standard errors clustered at the mosque level. All regressions include a misfit indicator. Outcome in columns (1)-(2) is the Women's Rights Index, an inverse covariance-weighted (ICW) index constructed following Anderson (2008) that aggregates: 1) allocation to a women's employment fund in a PKR 300 dictator game, 2) consent to having the respondent's wife independently surveyed by phone, and 3) willingness to sign a petition supporting women's political participation in Pakistani parliament. Outcome in columns (3)-(4) is the Neighbors' Rights Index which aggregates: 1) allocation to a poor neighbor fund in a PKR 300 dictator game, and 2) enumerator-rated community cleanliness on a 0–10 scale recorded independently before each survey. Both outcomes are in standard deviation units. Each column adds interactions between the pooled treatment indicators and one mosque-level characteristic: Columns (1) and (3) interact with mosque attendance tercile (low as base, interactions with medium and high tercile indicators); Columns (2) and (4) interact with residential/commercial status of the mosque. Randomization was stratified by both attendance tercile and residential/commercial status of the mosque.